

INDIGENE

STUDY CONSOLE

ANA 203

BASIC HISTOLOGY & CYTOLOGY

Complete Study Compendium — Expanded Edition

Federal University Oye-Ekiti | Nursing Science | 200 Level

The Cell • Membrane Transport • Cell Division • Epithelial Tissue • Connective Tissue • Muscle Tissue
Nerve Tissue • Blood • Cartilage • Bone • Nervous System Overview (Bonus)

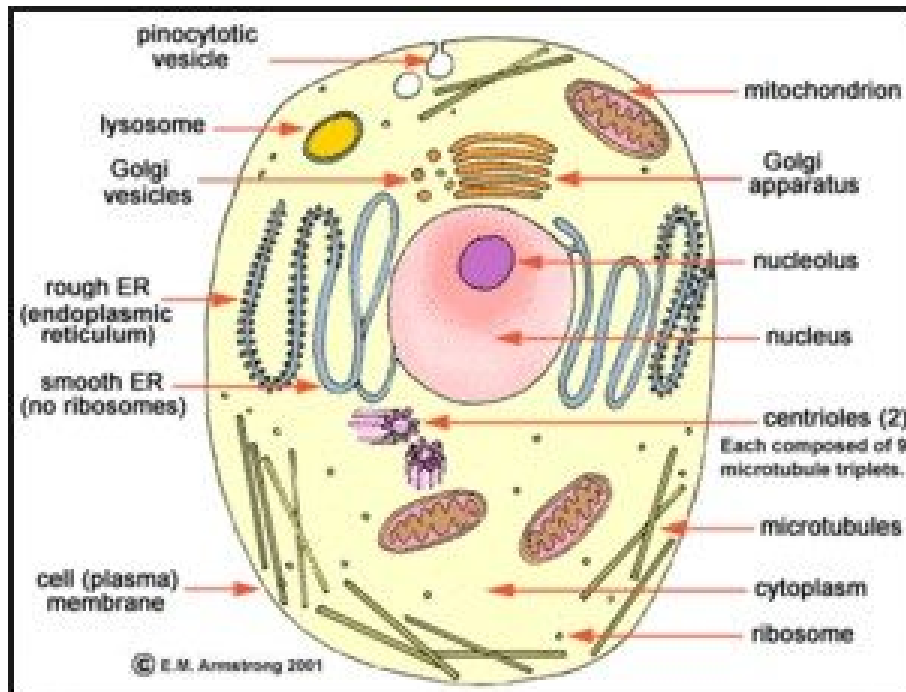
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■ What's new in this Expanded Edition

Every topic has been deepened with extra mechanism-level detail pulled from the fuller course reference material (ground substance chemistry, sarcomere/contraction mechanics, blood cell granule contents, bone matrix composition, and more) while keeping all original INDIGENE diagrams, Exam Hacks and Alzheimer's Mode blocks intact. A new bonus section on the Nervous System has been added at the end.

1. The Cell



Overview

- The cell is the smallest structural and functional unit of life — every tissue in the body is built from cells and their products. Cytology is the branch of science that studies cells.
- A generalized cell has four principal parts: (1) **Plasma membrane** — the outer limiting boundary; (2) **Cytoplasm** — the substance between nucleus and membrane; (3) **Organelles** — permanent, specialized structures; (4) **Inclusions** — temporary storage/secretion products (e.g. glycogen, melanin, lipid droplets).
- The clear fluid portion of cytoplasm surrounding the organelles is the **cytosol** — mainly protein, electrolytes, glucose, phospholipid, and esterified fatty acid.

Plasma (Cell) Membrane

- Described by the **Fluid Mosaic Model**: a phospholipid bilayer (~2.5 nm thick) with a monomolecular layer of protein, giving the membrane three functional layers overall.
- Phospholipids are amphipathic — hydrophilic heads face outward/inward toward water; hydrophobic tails face each other in the middle.
- Integral proteins span the membrane (channels, carriers); some form minute transport channels. Peripheral proteins sit on the surface (enzymes, receptors).
- Cholesterol stabilizes fluidity, preventing the membrane from becoming too rigid (cold) or too fluid (heat).
- Glycoproteins and glycolipids on the outer surface form the **glycocalyx** — enables cell recognition, acts as a binding site for other cells, provides receptor sites for hormones (e.g. insulin), contributes to immune recognition, and its negatively-charged carbohydrate moiety helps repel other negatively charged substances.

■ ALZHEIMER'S MODE

Picture the membrane as a lively dance floor (fluid) covered with mosaic tiles (proteins) — heads of the dancers point out toward the crowd (water), tails huddle together avoiding the crowd (hydrophobic core). Cholesterol is the bouncer keeping the dance floor from getting too wild or too stiff.

Organelles

Organelle	Structure	Function
Nucleus	Double membrane (envelope) with pores; contains nucleolus + chromatin	Control centre — stores DNA, directs protein synthesis
Rough ER	ER studded with ribosomes	Synthesizes/folds proteins for export/membranes; helps glycosylate proteins destined for the Golgi
Smooth ER	ER without ribosomes	Lipid/cholesterol synthesis, detox, calcium storage (muscle: sarcoplasmic reticulum)
Golgi apparatus	Cup-shaped stack of flattened, interconnected cisternae with vesicles at the borders	Modifies, packages, ships proteins/lipids; synthesizes carbohydrates the ER cannot (e.g. sialic acid, galactose); forms hyaluronic acid and chondroitin sulphate for mucus/glandular secretions
Mitochondria	Double membrane; inner folded into cristae; own DNA	'Power House' — ATP via aerobic respiration; matrix enzymes drive oxidative phosphorylation
Lysosomes	Membrane-bound sacs of hydrolytic enzymes	Intracellular digestion; 'suicide bags' in cell death
Peroxisomes	Small membrane-bound sacs	Break down fatty acids, detoxify hydrogen peroxide
Ribosomes	Free or ER-bound; rRNA + protein	Site of protein synthesis (translation)
Cytoskeleton	Microfilaments, microtubules, intermediate filaments	Shape, support, intracellular transport, movement
Centrioles	Paired cylindrical structures (9+0 triplets)	Organize the mitotic spindle during cell division

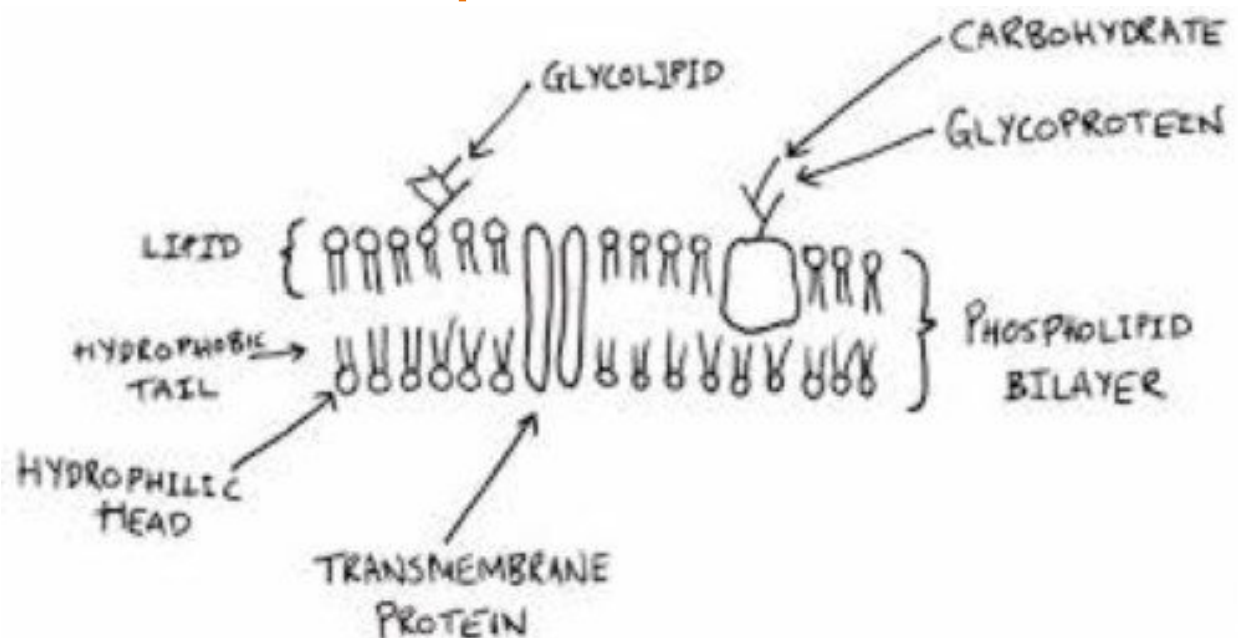
Nucleus (detail)

- Nuclear envelope: double membrane continuous with rough ER, perforated by nuclear pores controlling molecule traffic in/out.
- Nucleolus: dense region where ribosomal RNA is made and ribosome subunits assembled.
- Chromatin: DNA + histone proteins (delicate fibre network); condenses into visible chromosomes only during cell division.
- Chromosomes are threadlike structures made of units of inheritance called **genes**, in turn composed of DNA. Somatic cells contain 46 chromosomes in 23 pairs — the **diploid** number. Gametes (sperm/ova) carry only 23 — the **haploid** number.
- The 46 chromosomes comprise 44 **autosomes** (22 pairs) and 2 **sex chromosomes**.

■ **EXAM HACK — Organelle recall**

'Rough' ER = Ribosomes = protein factory. 'Smooth' ER = no ribosomes = lipids/detox. Mitochondria has its OWN DNA — a classic trick-question trap (endosymbiotic theory). Lysosome = 'Lysis' = breakdown/digestion; remember as the cell's stomach/demolition crew.

2. Membrane Transport



Passive Transport (no ATP required — moves WITH the concentration gradient)

Type	Description	Example
Simple diffusion	Small/non-polar molecules move directly across the lipid bilayer, high → low concentration	O ₂ , CO ₂ , lipid-soluble hormones
Facilitated diffusion	Molecules move through a channel or carrier protein, high → low, still no ATP	Glucose via GLUT transporters, ions via channels
Osmosis	Diffusion of WATER across a selectively permeable membrane toward higher solute concentration	Water movement in/out of RBCs
Filtration	Movement driven by hydrostatic (pressure) gradient, not concentration	Fluid filtration at the glomerulus/capillaries

Active Transport (REQUIRES ATP — moves AGAINST the gradient)

- **Primary active transport:** ATP is used directly by a pump protein, e.g. the Na⁺/K⁺-ATPase pump (3 Na⁺ out, 2 K⁺ in per ATP).
- **Secondary active transport:** uses the energy stored in an ion gradient (created by a primary pump) to move another substance — co-transport (symport) or counter-transport (antiport), e.g. Na⁺/glucose symporter in the gut.
- **Bulk transport** (vesicular, also ATP-dependent): Endocytosis — cell engulfs material (phagocytosis = 'cell eating' solids; pinocytosis = 'cell drinking' fluids; receptor-mediated endocytosis = specific ligand-receptor uptake). Exocytosis — vesicles fuse with the membrane to release contents outside (e.g. hormone/neurotransmitter secretion).

■ ALZHEIMER'S MODE

Passive transport is like water rolling downhill — no engine needed, gradient does the work. Active transport is like pushing a boulder uphill — you need fuel (ATP) because you're going against the natural flow.

■ EXAM HACK — Osmosis vs tonicity

Hypotonic solution → water rushes INTO the cell → cell swells/bursts (haemolysis in RBCs). Hypertonic solution → water rushes OUT → cell shrinks (crenation). Isotonic → no net water movement. Trap: 'facilitated diffusion needs ATP' is FALSE — it's still passive, just needs a protein escort, not fuel.

3. Cell Division — Mitosis & Meiosis

Mitosis (somatic cell division — growth & repair)

Phase	Key Event
Interphase (G1, S, G2)	Cell grows, DNA is replicated (S phase) — NOT technically part of mitosis proper
Prophase	Chromatin shortens/coils into chromosomes (sister chromatids); nucleoli and nuclear envelope become less distinct; centrioles separate to opposite poles; spindle forms
Metaphase	Centromeres of chromatid pairs line up on the equatorial plane; chromosomal microtubules attach centromeres to the spindle poles
Anaphase	Centromeres divide and identical sets of chromosomes move to opposite poles
Telophase	Nuclear membrane reappears and encloses chromosomes; chromosomes resume chromatin form; nucleoli reappear; spindle disappears; centrioles duplicate
Cytokinesis	Cytoplasm divides — two genetically identical diploid (2n) daughter cells result

Meiosis (gamete formation — reduces chromosome number by half)

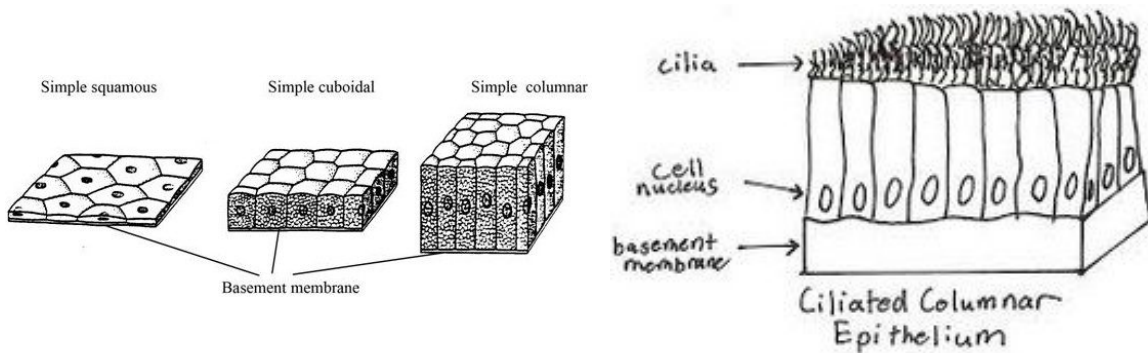
- Also called reduction division. Two successive divisions (Meiosis I and II) from one diploid cell, producing four genetically unique haploid (n) daughter cells.
- **Meiosis I** (reductional): homologous chromosomes pair up (synapsis) and exchange segments — **crossing over** — in Prophase I. Maternal and paternal chromosomes are then distributed to daughter cells entirely at random, shuffling genetic material further. Homologous pairs separate in Anaphase I.
- A short interphase follows Meiosis I (no further DNA replication).
- **Meiosis II** (equational): mechanically like mitosis — sister chromatids finally separate.
- Purpose: genetic variation (crossing over + independent assortment) and maintaining correct chromosome number across generations.

Feature	Mitosis	Meiosis
Divisions	One	Two (I and II)
Daughter cells	2, diploid (2n), identical	4, haploid (n), genetically unique
Purpose	Growth, repair, asexual reproduction	Gamete formation (sperm/egg)
Crossing over	No	Yes (Prophase I)

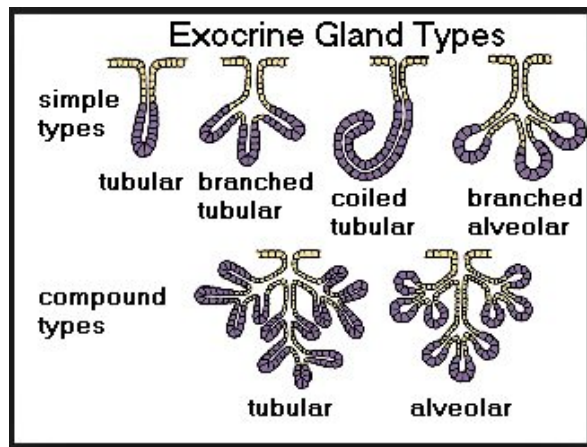
■ EXAM HACK — Mitosis vs Meiosis trap

'PMAT' for mitosis phase order — Prophase, Metaphase, Anaphase, Telophase. Meiosis = 'Meiosis makes 4' — four haploid daughter cells, NOT two. Crossing over happens ONLY in Prophase I of meiosis — never in mitosis; a favourite trick-question line.

4. Epithelial Tissue



	Simple	Stratified	
Squamous	 Simple squamous epithelium	 Stratified squamous epithelium	
Cuboidal	 Simple cuboidal epithelium	 Stratified cuboidal epithelium	
Columnar	 Simple columnar epithelium	 Stratified columnar epithelium	 Pseudostratified columnar epithelium



General Characteristics

- Derived from all three germ layers; covers body surfaces, lines cavities/organs/vessels, and forms glands.
- Cells are closely packed with minimal extracellular matrix, joined by junctions (tight, adherens, desmosomes, gap junctions).
- **Avascular** — no blood vessels (though nerves may extend through); nourished by diffusion from underlying connective tissue.
- Sits on a **basement membrane** (basal lamina + reticular lamina) that anchors it to underlying connective tissue and prevents it being torn away.
- Has an apical (free) surface and a basal surface — always polarized.
- Highly regenerative — damaged cells are replaced quickly and mitotic figures are common since epithelia are subjected to wear, tear and injury.
- All vital traffic of the body (digested food, O₂, waste, secretions) passes through epithelia; some epithelia are specialized to receive stimuli (sensory epithelium).

Classification by cell arrangement

- **Simple/unilayered** — one cell thick.
- **Pseudostratified** — appears layered but every cell touches the basement membrane.
- **Stratified/multilayered** — many cells thick.

Classification by cell shape

Type	Description	Example Location
Simple squamous	Single layer of flat, thin cells	Alveoli, capillary lining (endothelium), lining of body cavities (mesothelium)
Simple cuboidal	Single layer of cube-shaped cells, closely fitted polygons with central nucleus; secretion/absorption	Kidney tubules/collecting duct, thyroid, glandular ducts, ovary surface
Simple columnar	Single layer of tall, column-shaped cells; may bear microvilli/cilia	Lining of stomach, intestines (with goblet cells)
Pseudostratified columnar	Nuclei at different levels — looks layered; often ciliated (respiratory tract) or with goblet cells	Respiratory tract (trachea), upper respiratory tract, some male reproductive ducts
Stratified squamous	Multiple layers, flattened surface cells; keratinized (skin) or non-keratinized (moist surfaces)	Skin epidermis, oesophagus, mouth, tongue, vagina
Stratified cuboidal/columnar	Multiple layers of cuboidal/columnar cells (rare)	Sweat gland ducts, conjunctival fornix, part of male urethra, lactiferous ducts
Transitional	Outer cells large/rounded, drawn out into squamous-like cells when stretched	Urinary bladder, ureters, renal pelvis/calices, part of urethra

Glandular Epithelium

- **Exocrine glands:** secrete via ducts onto a surface (sweat, salivary, sebaceous glands; goblet cells) — have a secretory portion plus ducts that transport secretions out.
- **Endocrine glands:** ductless — secrete hormones directly into the bloodstream (e.g. thyroid, adrenal glands, pituitary, endocrine pancreas).
- **Structural classification of exocrine glands** (by shape of secretory part and branching of duct): simple (unbranched duct) vs compound (branched duct); further described as tubular, branched tubular (gastric/uterine glands), coiled tubular (sweat glands), acinar (seminal vesicle), branched acinar (sebaceous glands), or tubuloacinar (parotid, pancreas).
- **Exocrine secretion modes:** merocrine (exocytosis, cell stays intact — salivary glands, pancreas, goblet cells), apocrine (apical portion of cell pinches off — mammary glands), holocrine (whole cell ruptures and becomes the secretion — sebaceous glands).

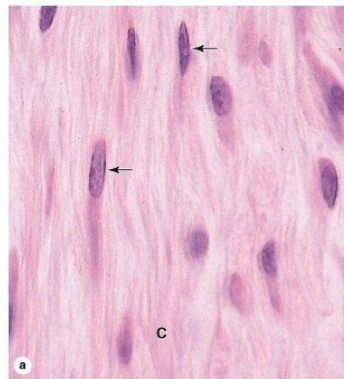
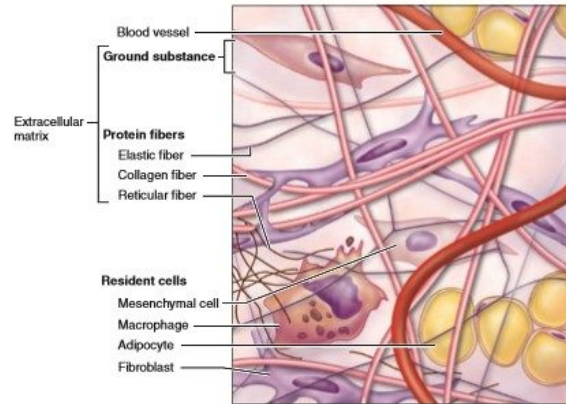
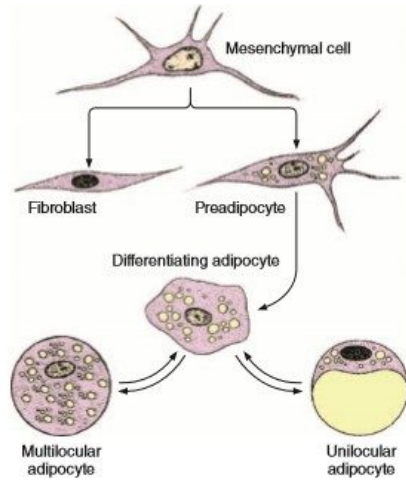
■ ALZHEIMER'S MODE

Think of epithelium as the body's 'wallpaper and wrapping paper' — it covers, lines, and wraps every free surface. Simple = one layer = built for fast exchange (thin walls). Stratified = many layers = built for protection (thick walls, like armour).

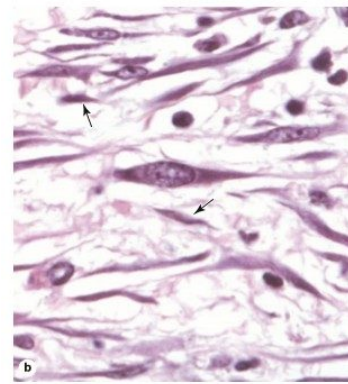
■ EXAM HACK — Classification trap

Naming rule: shape of the SURFACE (apical) layer names stratified epithelium, but for simple epithelium the shape names the WHOLE single layer. 'Pseudo' stratified is NOT truly layered — all cells touch the basement membrane, only nuclei are staggered. Transitional epithelium is the answer whenever a question mentions 'stretch' + 'bladder'.

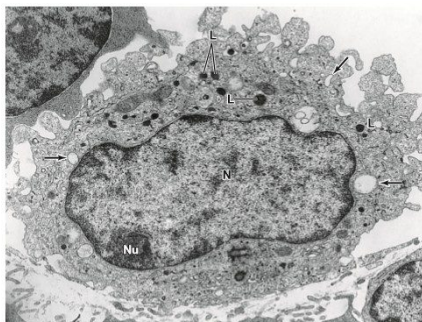
5. Connective Tissue Proper



(a) Fibroblasts typically have large active nuclei and eosinophilic cytoplasm that tapers off in both directions along the axis of the nucleus, a morphology often referred to as "spindle-shaped." Nuclei (arrows) are clearly seen, but the eosinophilic cytoplasmic processes resemble the collagen bundles (C) that fill the ECM and are difficult to distinguish in H&E-stained sections.



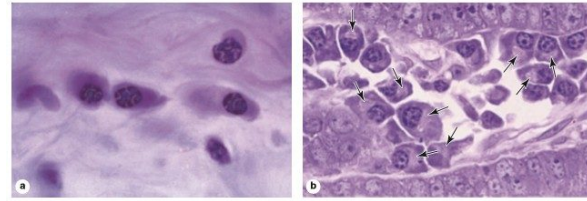
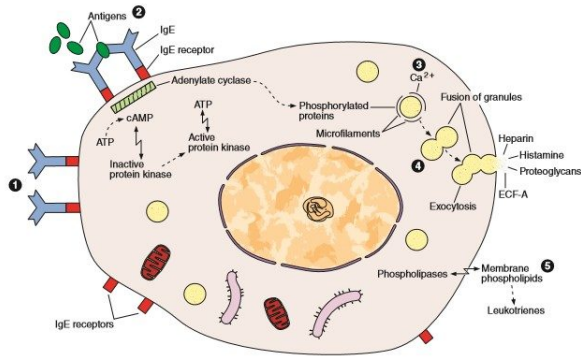
(b) Both active and quiescent fibroblasts may sometimes be distinguished, as in this section of dermis. Active fibroblasts have large, euchromatic nuclei and basophilic cytoplasm, while inactive fibroblasts (or fibrocytes) are smaller with more heterochromatic nuclei (arrows). The round, very basophilic round cells are leukocytes. Both X400. H&E.



Characteristic features of macrophages seen in this TEM of one such cell are the prominent nucleus (N) and the nucleolus (Nu) and the numerous secondary lysosomes (L). The arrows

indicate phagocytic vacuoles near the protrusions and indentations of the cell surface. X10,000.

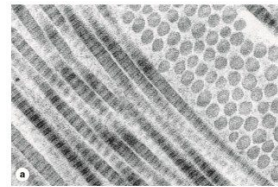
Cell Type	Major Locations	Main Function
Monocyte	Blood	Precursor of macrophages
Macrophage	Connective tissue, lymphoid organs, lungs, bone marrow, pleural and peritoneal cavities	Production of cytokines, chemotactic factors, and several other molecules that participate in inflammation (defense), antigen processing, and presentation
Kupffer cell	Liver (perisinusoidal)	Same as macrophages
Microglial cell	Central nervous system	Same as macrophages
Langerhans cell	Epidermis of skin	Antigen processing and presentation
Dendritic cell	Lymph nodes, spleen	Antigen processing and presentation
Osteoclast (from fusion of several macrophages)	Bone	Localized digestion of bone matrix
Multinuclear giant cell (several fused macrophages)	In connective tissue under various pathological conditions	Segregation and digestion of foreign bodies



Antibody-secreting plasma cells are present in variable numbers in the connective tissue of many organs.
(a) Plasma cells are large, ovoid cells, with basophilic cytoplasm. The round nuclei frequently show peripheral clumps of heterochromatin, giving the structure a "clock-face" appearance. X540. H&E.
(b) Plasma cells are often more abundant in infected tissues, as in the inflamed lamina propria shown here. A large pale

Golgi apparatus (arrows) at a juxtanuclear site in each cell is actively involved in the terminal glycosylation of the antibodies (glycoproteins). Plasma cells leave their sites of origin in lymphoid tissues, move to connective tissue, and produce antibodies that mediate immunity. X400 PT.

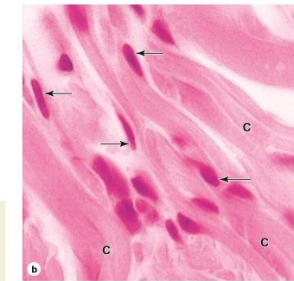
Type	Molecule Composition	Structure	Optical Microscopy	Major Locations	Main Function
Fibril-Forming Collagens					
I	[$\alpha 1(I)$] ₂ [$\alpha 2(I)$]	300-nm molecule, 67-nm banded fibrils	Thick, highly birefringent, fibers	Skin, tendon, bone, dentin	Resistance to tension
II	[$\alpha 1(II)$] ₃	300-nm molecule, 67-nm banded fibrils	Loose aggregates of fibrils, birefringent	Cartilage, vitreous body	Resistance to pressure
III	[$\alpha 1(III)$] ₃	67-nm banded fibrils	Thin, weakly birefringent, argyrophilic (silver-binding) fibers	Skin, muscle, blood vessels, frequently together with type I	Structural maintenance in expandable organs
V	[$\alpha 1(V)$] ₃	300-nm molecule, N-terminal globular domain	Frequently forms fiber together with type I	Fetal tissues, skin, bone, placenta, most interstitial tissues	Participates in type I collagen function
XI	[$\alpha 1(XI)$] ₃ [$\alpha 2(XI)$] ₃	300-nm molecule	Small fibers	Cartilage	Participates in type II collagen function
Sheet-Forming Collagens					
IV	[$\alpha 1(IV)$] ₃ [$\alpha 2(IV)$] ₃	2-dimensional cross-linked network	Detected by immunocytochemistry	All basal and external laminae	Support of epithelial cells; filtration
Linking/Anchoring Collagens					
VII	[$\alpha 1(VII)$] ₃	450-nm, globular domain at each end	Detected by immunocytochemistry	Epithelial basement membranes	Anchors basal laminae to underlying reticular lamina
IX	[$\alpha 1(IX)$] ₃ [$\alpha 2(IX)$] ₃	200-nm molecule	Detected by immunocytochemistry	Cartilage, vitreous body	Binds various proteoglycans; associated with type II collagen
XII	[$\alpha 1(XII)$] ₃	Large N-terminal domain	Detected by immunocytochemistry	Placenta, skin, tendons	Interacts with type I collagen
XIV	[$\alpha 1(XIV)$] ₃	Large N-terminal domain; cross-shaped molecule	Detected by immunocytochemistry	Placenta, bone	Binds type I collagen fibrils, with types V and XII, strengthening fiber formation



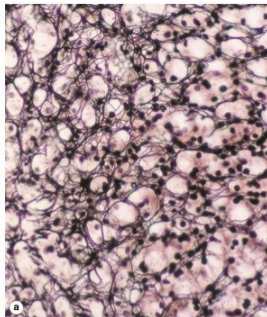
Subunits of type I collagen, the most abundant collagen, assemble to form extremely strong fibrils, which are then bundled together further by other collagens into much larger structures called collagen fibers.

(a) TEM shows fibrils cut longitudinally and transversely. In longitudinal sections fibrils display alternating dark and light bands; in cross section the cut ends of individual collagen molecules appear as dots. Ground substance completely surrounds the fibrils. X100,000.

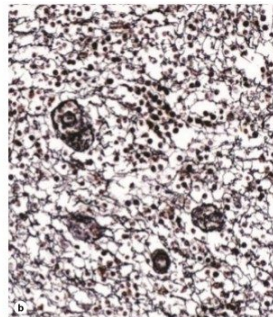
(b) The large bundles of type I collagen fibrils (C) appear as acidophilic collagen fibers in connective tissues, where they



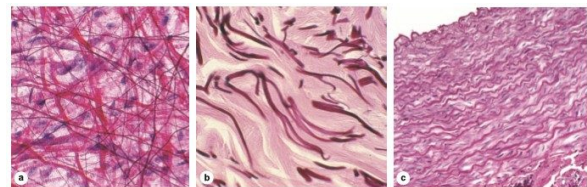
may fill the extracellular space. Subunits for these fibers were secreted by the fibroblasts (arrows) associated with them. X400. H&E.



In these silver-stained sections of adrenal cortex (a) and lymph node (b), networks of delicate, black reticular fibers are prominent. These fibers serve as a supportive stroma in most lymphoid and hematopoietic organs and many endocrine



glands. The fibers consist of type III collagen that is heavily glycosylated, producing the black argyrophilia. Cell nuclei are also dark, but cytoplasm is unstained. X100.



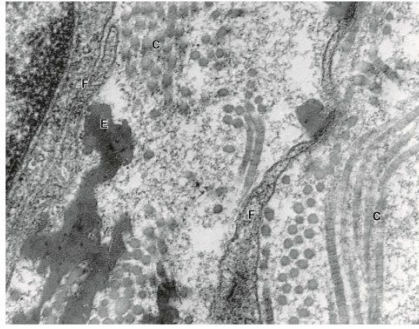
Elastic fibers or lamellae (sheets) add resiliency to connective tissue. Such fibers may be difficult to discern in H&E-stained tissue, but elastin has a distinct, darker-staining appearance with other staining procedures.

(a) The length, diameter, distribution, and density of dark elastic fibers are easily seen in this spread preparation of nonstretched connective tissue in a mesentery. X200. Hematoxylin and orcein.

(b) In sectioned tissue at higher magnification, elastic fibers can be seen among the acidophilic collagen bundles of dermis. X400. Aldehyde fuchsin.

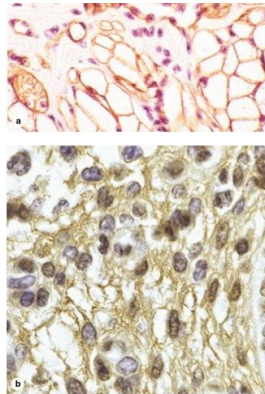
(c) Elastic lamellae in the wall of the aorta are more darkly stained, incomplete sheets of elastin between the layers of eosinophilic smooth muscle. X90. H&E.

FIGURE 15-15 Ultrastructure of the extracellular matrix (ECM).



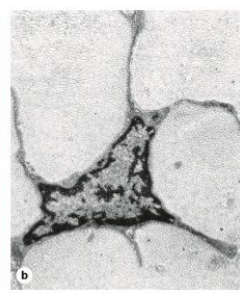
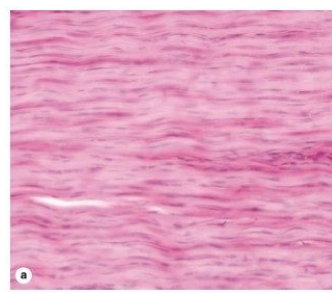
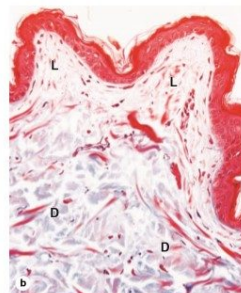
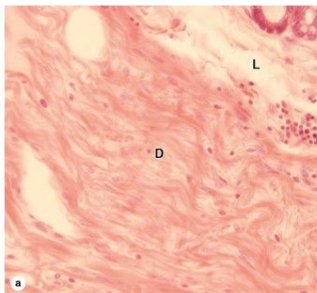
TEM of the connective tissue ECM reveals ground substance as either empty or containing fine granular material that fills spaces between the collagen (C) and elastic (E) fibers and surrounds fibroblast cells and processes (F). The granularity of ground substance is an artifact of the glutaraldehyde-tannic acid fixation procedure. X100,000.

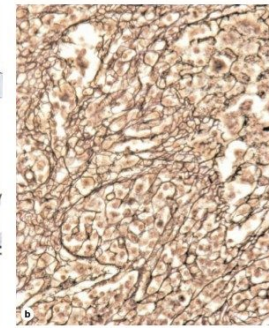
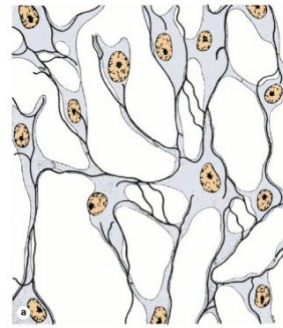
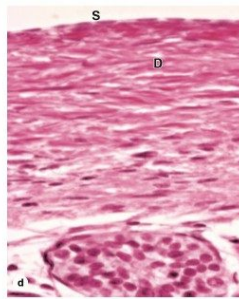
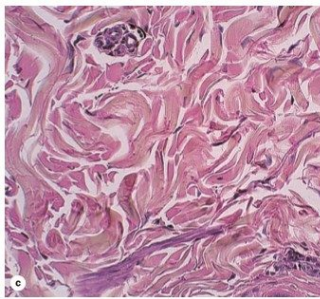
Glycosaminoglycan	Repeating Disaccharides		Distribution	Electrostatic Interaction with Collagen
	Hexuronic Acid	Hexosamine		
Hyaluronic acid	D-glucuronic acid	D-glucosamine	Umbilical cord, synovial fluid, vitreous humor, cartilage	
Chondroitin 4-sulfate	D-glucuronic acid	D-galactosamine	Cartilage, bone, cornea, skin, notochord, aorta	High levels of interaction, mainly with collagen type II
Chondroitin 6-sulfate	D-glucuronic acid	D-galactosamine	Cartilage, umbilical cord, skin, aorta (media)	High levels of interaction, mainly with collagen type II
Dermatan sulfate	L-iduronic acid or D-glucuronic acid	D-galactosamine	Skin, tendon, aorta (adventitia)	Low levels of interaction, mainly with collagen type I
Heparan sulfate	D-glucuronic acid or L-iduronic acid	D-galactosamine	Aorta, lung, liver, basal laminae	Intermediate levels of interaction, mainly with collagen types III and IV
Keratan sulfate	D-galactose	D-glucosamine	Cartilage, nucleus pulposus, annulus fibrosus	None



Both of these glycoproteins (and other similar glycoproteins) are multibinding, with binding sites for ECM components and for integrins at cell surfaces, and have important roles in cell migration and maintaining tissue structure.
 (a) **Laminin** is concentrated in the basal lamina of the stratified epithelium (top) and in the external laminae surrounding cross-sectioned nerves and muscle fibers. X200.
 (b) A fine network of **fibronectin** is localized more diffusely throughout the ECM. X400.

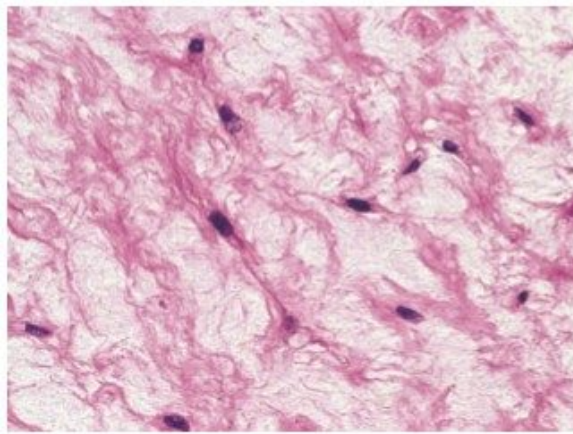
Classification of connective or supporting tissues.			
	General Organization	Major Functions	Examples
Connective Tissue Proper			
Loose (areolar) connective tissue	Much ground substance; many cells and little collagen, randomly distributed	Supports microvasculature, nerves, and immune defense cells	Lamina propria beneath epithelial lining of digestive tract
Dense irregular connective tissue	Little ground substance; few cells (mostly fibroblasts); much collagen in randomly arranged fibers	Protects and supports organs; resists tearing	Dermis of skin, organ capsules, submucosa layer of digestive tract
Dense regular connective tissue	Almost completely filled with parallel bundles of collagen; few fibroblasts, aligned with collagen	Provide strong connections within musculoskeletal system; strong resistance to force	Ligaments, tendons, aponeuroses, corneal stroma
Embryonic Connective Tissues			
Mesenchyme	Sparse, undifferentiated cells, uniformly distributed in matrix with sparse collagen fibers	Contains stem/progenitor cells for all adult connective tissue cells	Mesodermal layer of early embryo
Mucoid (mucous) connective tissue	Random fibroblasts and collagen fibers in viscous matrix	Supports and cushions large blood vessels	Matrix of the fetal umbilical cord
Specialized Connective Tissues			
Reticular connective tissue	Delicate network of reticulin/collagen III with attached fibroblasts (reticular cells)	Supports blood-forming cells, many secretory cells, and lymphocytes in most lymphoid organs	Bone marrow, liver, pancreas, adrenal glands, all lymphoid organs except the thymus
Adipose Tissue			
Cartilage			
Bone			
Blood			





(a) The diagram shows only the fibers and attached reticular cells (free, transient cells are not represented). Reticular fibers of type III collagen are produced and enveloped by the reticular cells, forming an elaborate network through which interstitial fluid or lymph and wandering cells from blood pass continuously.

(b) The micrograph shows a silver-stained section of lymph node in which reticular fibers are seen as irregular black lines. Reticular cells are also heavily stained and dark. Most of the smaller, more lightly stained cells are lymphocytes passing through the lymph node. X200. Silver.



General Characteristics

- Most abundant and widely distributed tissue — connects, supports, protects, and binds other tissues together; provides a matrix that physically connects tissues/cells, and its interstitial fluid gives metabolic support as the medium for diffusion of nutrients and waste.
- Unlike epithelium: cells are sparse and widely spaced within an abundant **extracellular matrix (ECM)** of protein fibres (collagen, elastic) plus ground substance.
- Usually well **vascularized** (except cartilage, which is avascular).
- Derived embryologically from **mesenchyme** (mesoderm) — mesenchymal cells are undifferentiated, spindle-shaped, with large euchromatic nuclei; besides connective tissue proper, bone and cartilage, mesenchyme also supplies stem cells for blood, vascular endothelium, and muscle.

Cell Types

Cell	Function
Fibroblasts	Most abundant resident cell (permanent, locally derived from mesenchyme); synthesize collagen, elastin, GAGs, proteoglycans and multiadhesive glycoproteins of the matrix. 'Fibrocyte' is sometimes used for the quiescent form
Macrophages	Phagocytose debris/pathogens; derived from monocytes (mononuclear phagocyte system); organ-specific names include Kupffer cells (liver), microglia (CNS), Langerhans cells (skin), osteoclasts (bone)

Cell	Function
Mast cells	Release histamine/heparin — mediate inflammation and allergic (type I hypersensitivity) response; granules show metachromasia with toluidine blue
Plasma cells	Derived from B-lymphocytes; secrete antibodies (immunoglobulins); RER-rich basophilic cytoplasm, eccentric 'clock-face' nucleus
Adipocytes	Store fat (triglycerides); energy reserve and insulation
Leukocytes	White blood cells that migrate into tissue during inflammation via diapedesis

Fibres of the Matrix

- **Collagen fibres** — thick, strong, flexible but resistant to stretch; the most abundant protein in the body (~30% of dry weight). Type I forms large eosinophilic bundles (tendons, capsules, dermis); Type III (reticular) forms fine networks; Type IV forms basement membrane sheets.
- **Elastic fibres** (elastin) — thin, branching, allow stretch and recoil (blood vessel walls, lungs); built from fibrillin microfibrils scaffolding elastin deposition; cross-linked by desmosine bonds.
- **Reticular fibres** — fine type III collagen forming delicate supportive networks/stroma for soft organs (liver, spleen, lymph nodes, bone marrow); silver-stainable (argyrophilic).

Ground Substance

- A highly hydrated, viscous mixture of **glycosaminoglycans (GAGs)**, **proteoglycans**, and **multiadhesive glycoproteins** — fills space between cells/fibres, lubricates, and acts as a barrier to invaders.
- Hyaluronic acid is the largest, most ubiquitous GAG (unsulfated); other GAGs (dermatan, chondroitin, keratan, heparan sulfate) are smaller and sulfated, covalently bound to core proteins as proteoglycans.
- Multiadhesive glycoproteins — laminin (epithelial adhesion) and fibronectin (fibroblast-derived, binds collagen/GAGs, guides cell migration) — bind cell-surface integrin receptors, physically linking the ECM to the cytoskeleton.

Classification

Type	Description	Example
Loose (areolar)	Cells, fibres and ground substance in roughly equal parts; delicate, flexible, not very stress-resistant	Beneath skin, around organs/vessels, lamina propria of the gut
Dense regular	Densely packed, parallel collagen bundles with fibroblasts aligned for one-directional stress	Tendons, ligaments
Dense irregular	Densely packed collagen running in multiple/random directions — resists stress from all directions	Dermis of skin, organ capsules
Adipose	Cells dominated by fat storage (signet-ring appearance)	Subcutaneous fat, around kidneys
Reticular	Fine reticular fibre meshwork (produced by reticular cells) supporting free cells	Lymph nodes, spleen, bone marrow stroma
Mucoid (mucous)	Embryonic type; jellylike, hyaluronic-acid-rich matrix with sparse collagen and scattered fibroblasts	Umbilical cord (Wharton's jelly), pulp of young teeth

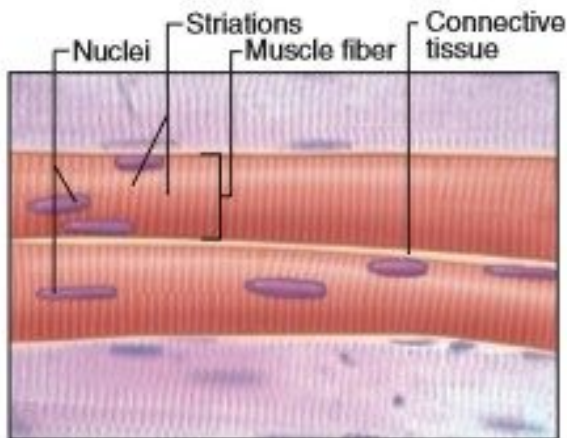
■ MEDICAL APPLICATION

A **keloid** is abnormal excess collagen forming in skin scars — more common in individuals of African descent and prone to recur after excision. Mutations in fibrillin (elastic fibre scaffold protein) cause **Marfan syndrome**, weakening elastic-rich tissue and predisposing to aortic aneurysm.

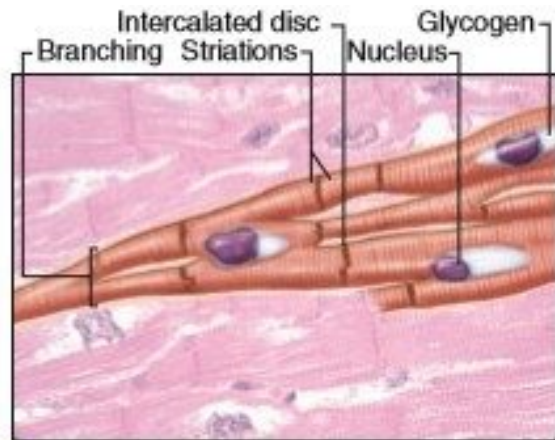
■ EXAM HACK — Fibre recall

Collagen = ROPE (strong, doesn't stretch). Elastin = RUBBER BAND (stretches and snaps back). Reticular = NET (fine supportive mesh for soft organs). Common trap: calling adipose tissue 'just fat storage' — remember it's a true connective tissue with cells (adipocytes) in a matrix.

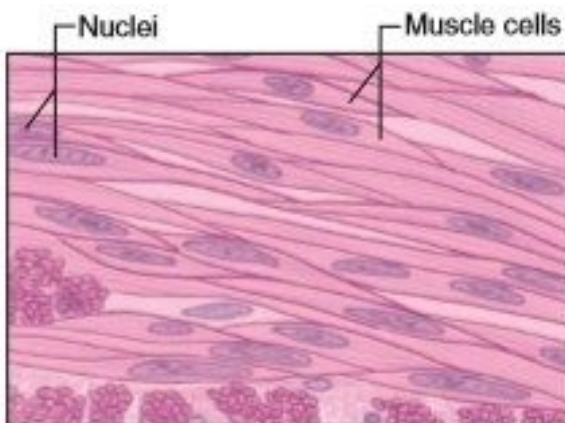
6. Muscle Tissue



● Skeletal muscle



● Cardiac muscle



● Smooth muscle

>> MEDICAL APPLICATION

The variation in diameter of muscle fibers depends on factors such as the specific muscle, age, gender, nutritional status, and physical training of the individual. Exercise enlarges the skeletal musculature by stimulating formation of new myofibrils and growth in the diameter of individual muscle fibers. This process, characterized by increased cell volume, is called **hypertrophy** (Gr. *hyper*, above + *trophe*, nourishment). Tissue growth by an increase in the number of cells is termed **hyperplasia** (*hyper* + Gr. *plasis*, molding), which takes place very readily in smooth muscle, whose cells have not lost the capacity to divide by mitosis.

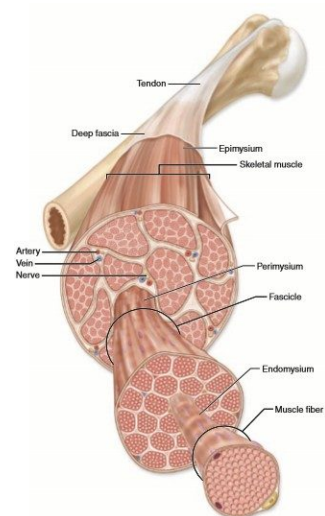
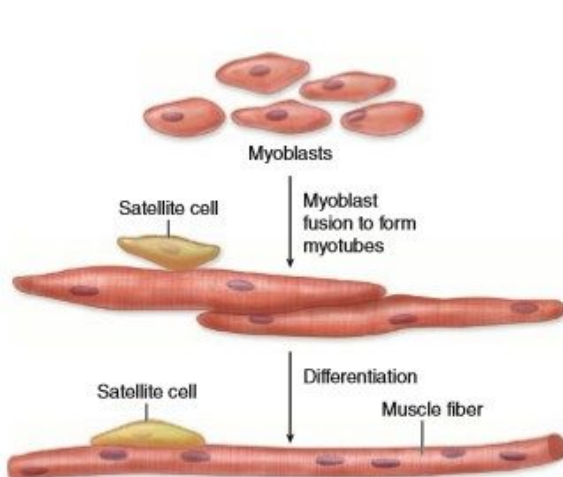
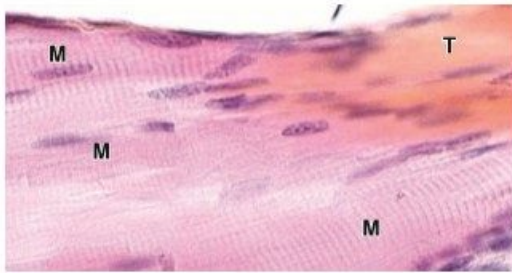
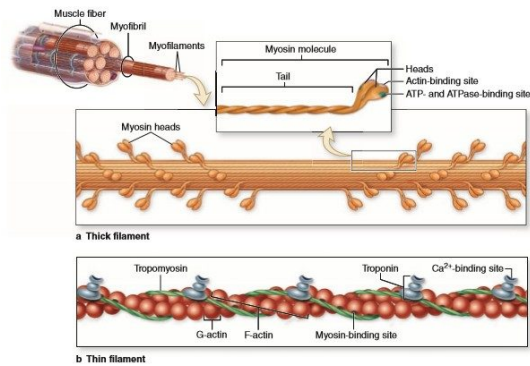
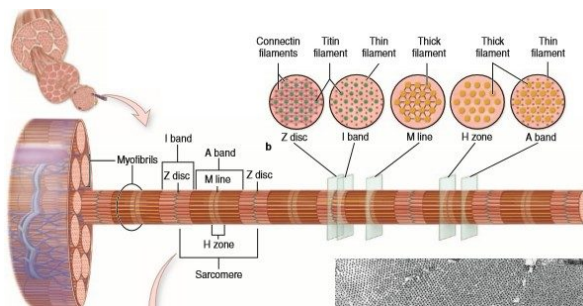
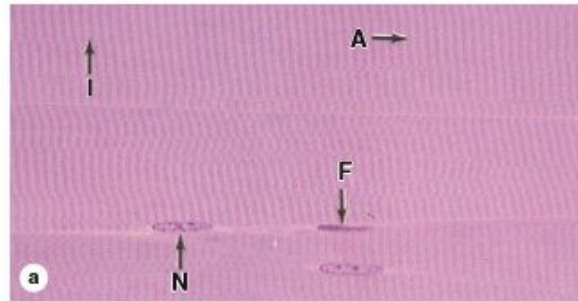


FIGURE 10-10 Myotendinous junction.

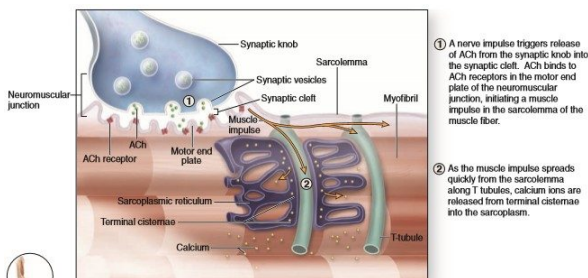


Tendons develop together with skeletal muscles and join muscles to the periosteum of bones. The collagen fibers of a tendon (T) are continuous with those in the connective tissue layers around muscle fibers (M), forming a strong unit that allows muscle contraction to move the skeleton. X400. H&E.



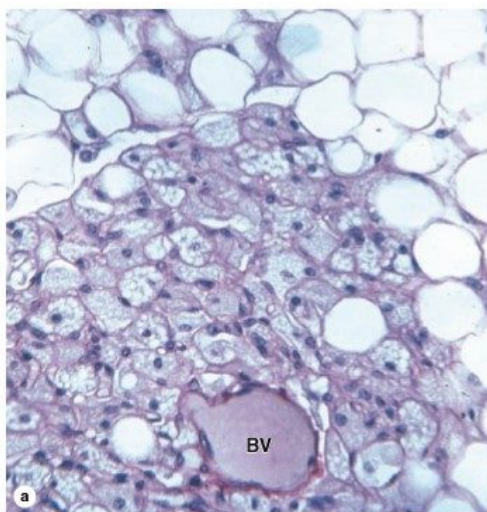
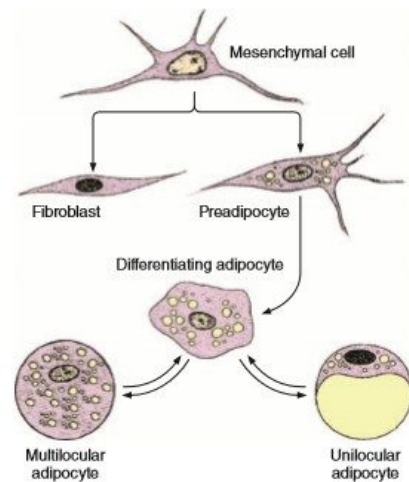
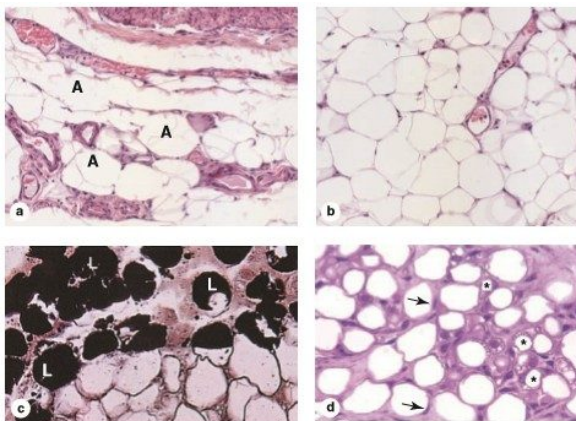
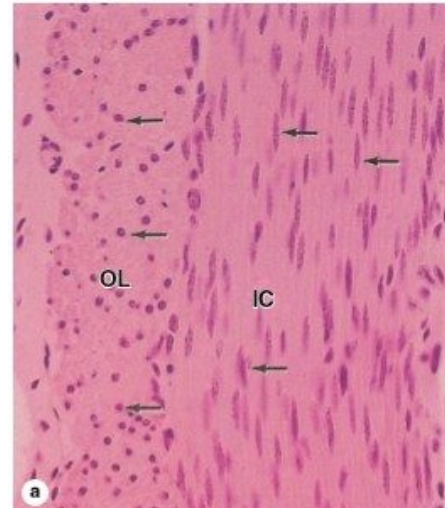
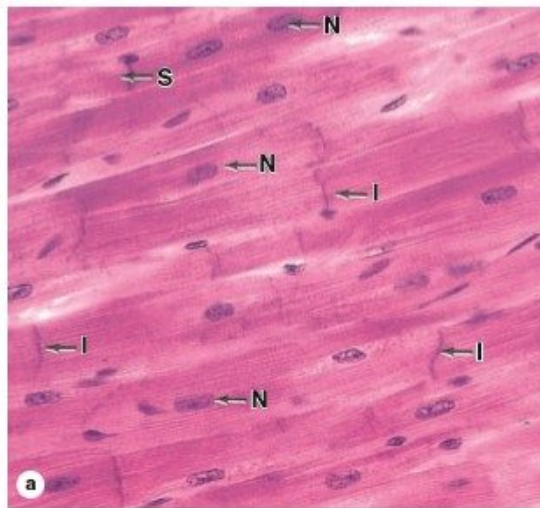
Myofilaments, which include both thick and thin filaments, consist of contractile protein arrays bundled within myofibrils.

(a) A thick myofilament contains 200-500 molecules of **myosin**. (b) A thin filament contains **F-actin**, **tropomyosin**, and **troponin**.



- ① A nerve impulse triggers release of ACh from the synaptic knob into the synaptic cleft. ACh binds to ACh receptors in the motor end plate of the neuromuscular junction, initiating a muscle impulse in the sarcolemma of the muscle fiber.
- ② As the muscle impulse spreads quickly from the sarcolemma along T tubules, calcium ions are released from terminal cisternae into the sarcoplasm.

TABLE 10-1	Major characteristics of skeletal muscle fiber types.		
	Slow, Oxidative Fibers (Type I)	Fast, Oxidative-Glycolytic Fibers (Type IIa)	Fast, Glycolytic Fibers (Type IIb)
Mitochondria	Numerous	Numerous	Sparse
Capillaries	Numerous	Numerous	Sparse
Fiber diameter	Small	Intermediate	Large
Size of motor unit	Small	Intermediate	Large
Myoglobin content	High (red fibers)	High (red fibers)	Low (white fibers)
Glycogen content	Low	Intermediate	High
Major source of ATP	Oxidative phosphorylation	Oxidative phosphorylation	Anaerobic glycolysis
Glycolytic enzyme activity	Low	Intermediate	High
Rate of fatigue	Slow	Intermediate	Fast
Myosin-ATPase activity	Low	High	High
Speed of contraction	Slow	Fast	Fast
Typical major locations	Postural muscles of back	Major muscles of legs	Extraocular muscles



General Characteristics

- Specialized for contraction — generates force and movement; virtually all muscle cells are mesodermal in origin.
- Key properties: excitability (responds to stimuli), contractility (shortens), extensibility (can be stretched), elasticity (recoils).
- Composed of elongated cells/fibres containing contractile proteins actin and myosin. Muscle-specific terms: sarcoplasm (cytoplasm), sarcoplasmic reticulum (smooth ER), sarcolemma (cell membrane + external lamina).

Feature	Skeletal	Cardiac	Smooth
Appearance	Striated	Striated	Non-striated
Control	Voluntary	Involuntary	Involuntary
Nuclei	Multinucleate (peripheral)	Single/binucleate (central)	Single (central)
Cell shape	Long, cylindrical fibres	Branched, short cells	Spindle-shaped, tapered
Special features	Attached to bone via tendons; myotendinous junctions	Intercalated discs join cells; gap junctions	Found in walls of hollow organs/vessels

Skeletal Muscle — Organization

- Fibres are long, cylindrical, multinucleated cells (10-100 μm diameter), formed embryonically when myoblasts fuse into myotubes. A reserve population of **satellite cells** beneath the sarcolemma proliferates to regenerate fibres after injury.
- Connective tissue wrapping: **epimysium** (whole muscle) → **perimysium** (each fascicle/bundle) → **endomysium** (individual fibre, rich capillary network). These layers transmit contractile force and continue into tendons at myotendinous junctions.
- Cross-striations: dark **A bands** (anisotropic, thick myosin filaments) alternate with light **I bands** (isotropic, thin actin filaments), bisected by the **Z disc**. The **sarcomere** (Z disc to Z disc, $\sim 2.5 \mu\text{m}$) is the contractile unit.
- Thin filaments carry regulatory proteins **tropomyosin** (blocks myosin-binding sites at rest) and **troponin** (TnT binds tropomyosin, TnC binds Ca^{2+} , TnI regulates the actin-myosin interaction).

Mechanism of Contraction

- A nerve impulse at the **neuromuscular junction (motor end plate)** travels along T tubules to the sarcoplasmic reticulum, releasing Ca^{2+} .
- Ca^{2+} binds troponin, shifting tropomyosin to expose myosin-binding sites on actin, allowing crossbridge formation — the **sliding filament** mechanism (filaments themselves don't shorten; they slide past each other).
- When the impulse ends, Ca^{2+} is pumped back into the sarcoplasmic reticulum, tropomyosin re-blocks the sites, and the sarcomere relaxes.
- Without ATP, actin-myosin crossbridges stay locked — the basis of *rigor mortis* after death.

■ MEDICAL APPLICATION

Myasthenia gravis is an autoimmune disorder with antibodies against acetylcholine receptors at the neuromuscular junction, causing intermittent skeletal muscle weakness (often first affecting the eye muscles).

Cardiac and Smooth Muscle

- Cardiac muscle cells have abundant mitochondria (~40% of cytoplasmic volume) for continuous aerobic work; **intercalated discs** contain desmosomes/fascia adherentes (mechanical coupling) and gap junctions (electrical coupling, allowing the heart to act as a functional syncytium). Cardiac muscle lacks satellite cells and has little regenerative capacity.
- Smooth muscle lacks troponin — contraction instead uses calmodulin and Ca^{2+} -sensitive myosin light-chain kinase. It regenerates rapidly after injury because its cells are small and less differentiated.

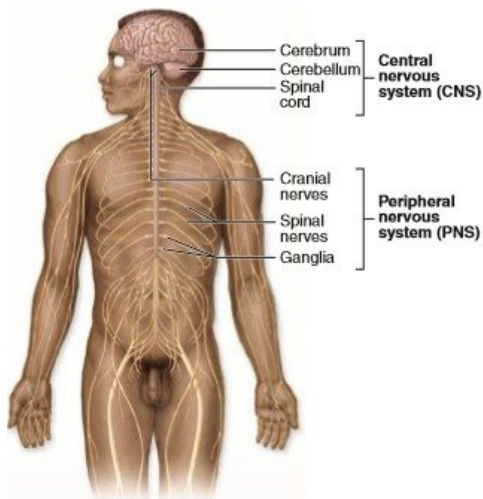
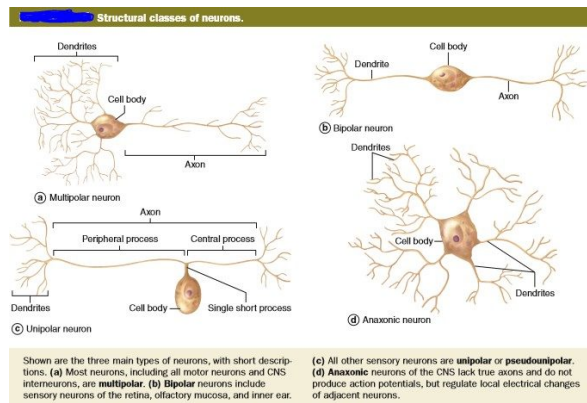
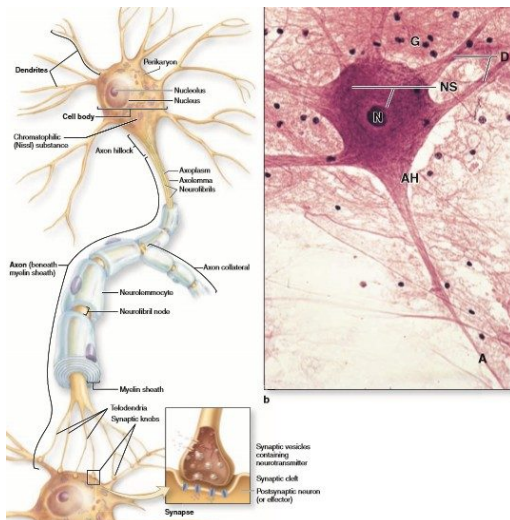
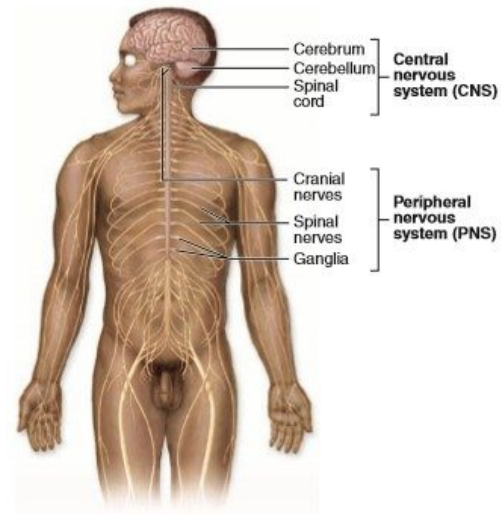
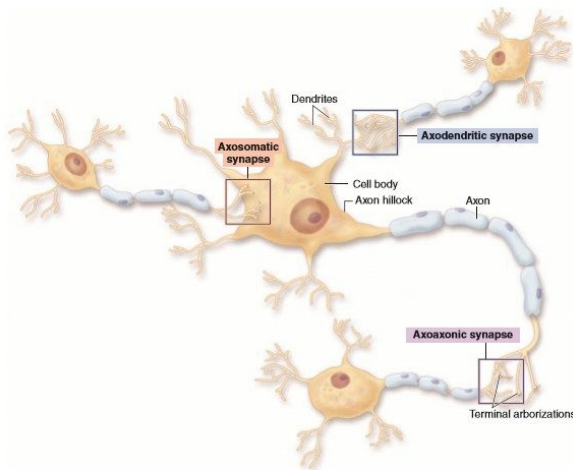
Clinical Correlations

- Skeletal muscle damage can raise serum creatine kinase (CK) — used to detect muscle injury/myocardial infarction.
- Disruption of intercalated discs contributes to cardiac arrhythmias.
- Smooth muscle dysfunction is implicated in hypertension (vascular smooth muscle) and irritable bowel syndrome (GI smooth muscle).
- Benign smooth-muscle tumours (leiomyomas/'fibroids') commonly develop in the uterine wall.

■ EXAM HACK — Muscle ID trap

'Striated + voluntary' = Skeletal. 'Striated + involuntary' = Cardiac (intercalated discs are the giveaway word).
'Non-striated + involuntary' = Smooth. Multinucleate + peripheral nuclei is ALWAYS skeletal muscle — a classic image-ID question.

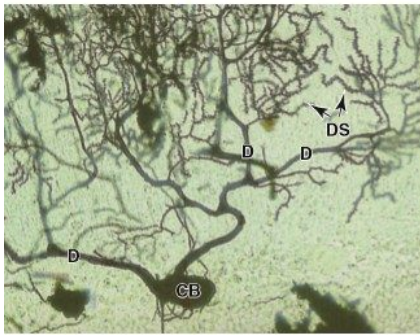
7. Nerve Tissue



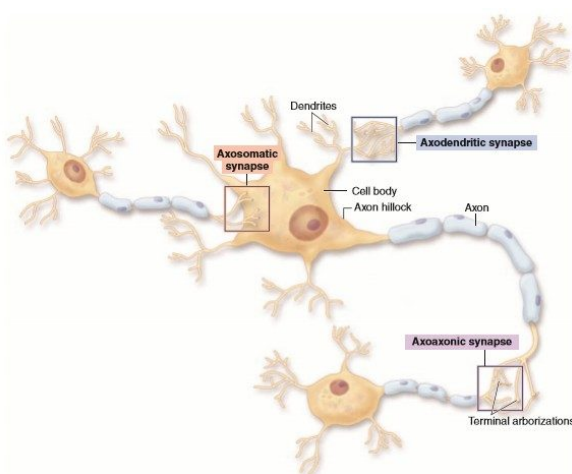
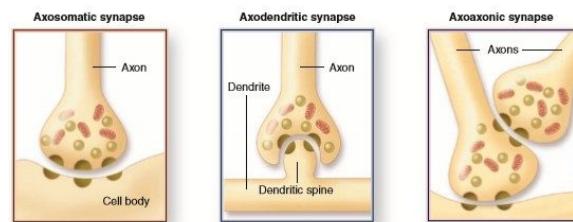
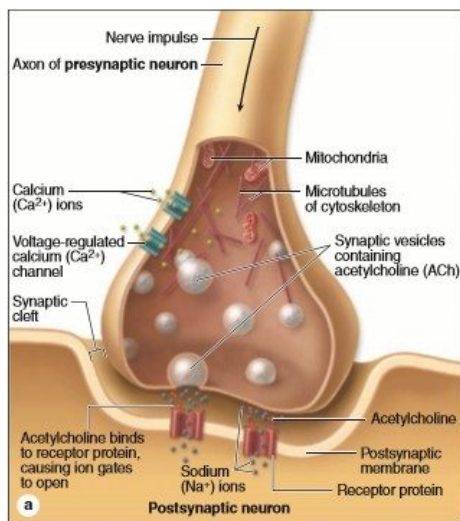
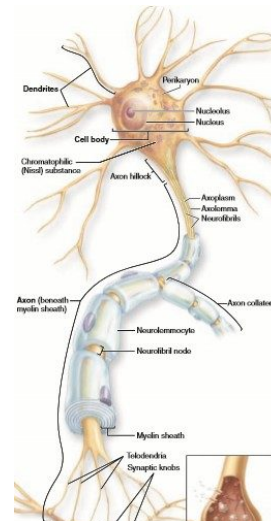
>> MEDICAL APPLICATION

Parkinson disease is a slowly progressing disorder affecting muscular activity characterized by tremors, reduced activity of the facial muscles, loss of balance, and postural stiffness. It is caused by gradual loss by apoptosis of dopamine-producing neurons whose cell bodies lie within the nuclei of the CNS substantia nigra. Parkinson disease is treated with L-dopa (L-3,4-dihydroxyphenylalanine), a precursor of dopamine which augments the declining production of this neurotransmitter.

9-5 Dendrites and dendritic spines.



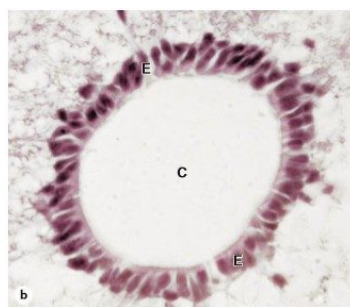
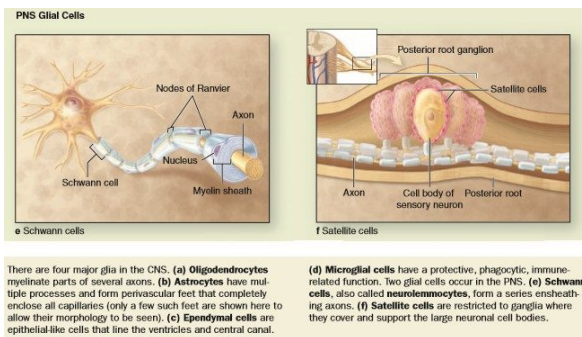
The large Purkinje neuron in this silver-impregnated section of cerebellum has many dendrites (D) emerging from its cell body (CB) and forming branches. The small dendritic branches have many short projecting dendritic spines (DS) spaced closely along their length, each of which is a site of a synapse with another neuron. X650. Silver stain.



Common neurotransmitters and their actions.	
Neurotransmitter	Description/Action
ACETYLCHOLINE (ACh)	
$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{N}^+-\text{CH}_2-\text{CH}_2-\text{O}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	Chemical structure significantly different from that of other neurotransmitters; active in CNS and in both somatic and autonomic parts of PNS; binds to ACh receptors (cholinergic receptors) in PNS to open ion channels in postsynaptic membrane and stimulate muscle contraction
AMINO ACIDS	
$\begin{array}{c} \text{O} \\ \\ \text{NH}_2-\text{CH}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$	Molecules with both carboxyl ($-\text{COOH}$) and amine ($-\text{NH}_2$) groups and various R groups; act as important transmitters in the CNS
Glutamate	Excites activity in neurons to promote cognitive function in the brain (learning and memory); most common neurotransmitter in the brain; opens Na^+ channels
Gamma-aminobutyric acid (GABA)	Synthesized from glutamate; primary inhibitory neurotransmitter in the brain; also influences muscle tone; opens or closes various ion channels
Glycine	Inhibits activity between neurons in the CNS, including retina; opens Cl^- channels
MONOAMINES	
$\begin{array}{c} \text{OH} \\ \\ \text{NH}_2-\text{CH}_2-\text{CH}-\text{C}_6\text{H}_3(\text{OH})_2 \end{array}$	Molecules synthesized from an amino acid by removing the carboxyl group and retaining the single amine group; also called biogenic amines
Serotonin or 5-hydroxytryptamine (5-HT)	Has various functions in the brain related to sleep, appetite, cognition (learning, memory), and mood; modulates actions of other neurotransmitters
Catecholamines	A distinct group of monoamines
Dopamine	Produces inhibitory activity in the brain; important roles in cognition (learning, memory), motivation, behavior, and mood; opens K^+ channels, closes Ca^{2+} channels
Norepinephrine (noradrenaline)	Neurotransmitter of PNS (sympathetic division of autonomic nervous system) and specific CNS regions

NEUROPEPTIDES	
	Small polypeptides act as signals to assist in and modulate communication among neurons in the CNS
Enkephalin	Helps regulate response to noxious and potentially harmful stimuli
Neuropeptide Y	Involved in memory regulation and energy balance (increased food intake and decreased physical activity)
Somatostatin	Inhibits activities of neurons in specific brain areas
Substance P	Assists with pain information transmission into the brain
Cholecystokinin (CCK)	Stimulates neurons in the brain to help mediate satiation (fullness) and repress hunger
Beta-endorphin	Prevents release of pain signals from neurons and fosters a feeling of well-being
Neurotensin	Helps control and moderate the effects of dopamine
OTHERS	
Adenosine	Also part of a nucleotide, inhibits activities in certain CNS neurons
Nitric oxide	Involved in learning and memory; relaxes muscle in the digestive tract; important for relaxation of smooth muscle in blood vessels (vasodilation)

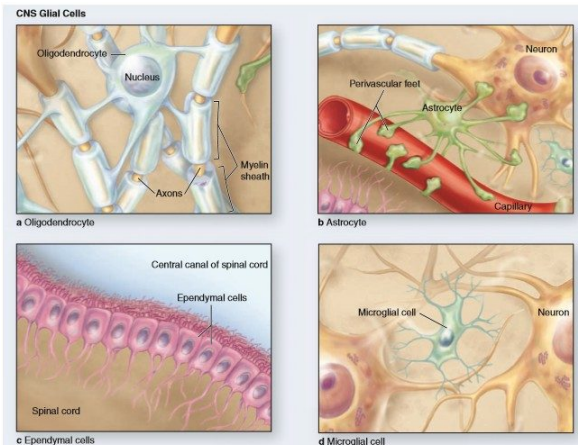
Glial Cell Type	Origin	Location	Main Functions
Oligodendrocyte	Neural tube	CNS	Myelin production, electrical insulation
Schwann cell (Neurolemmocyte)	Neural crest	Peripheral nerves	Myelin production, electrical insulation
Astrocyte	Neural tube	CNS	Structural and metabolic support of neurons; BBB; repair processes
Satellite cells (of ganglia)	Neural crest	Peripheral ganglia	Structural and metabolic support for neuronal cell bodies
Ependymal cell	Neural tube	Line ventricles and central canal of CNS	Aid production and movement of CSF
Microglia	Bone marrow (monocytes)	CNS	Defense and immune-related activities



Ependymal cells are epithelial-like cells that form a single layer lining the fluid-filled ventricles and central canal of the CNS. (a) Lining the ventricles of the cerebrum, columnar ependymal cells (E) extend cilia and microvilli from the apical surfaces into the ventricle (V). These modifications help circulate the CSF and monitor its contents. Ependymal cells have junctional complexes at their apical ends like those of epithelial cells but lack a basal lamina. The cells' basal ends are tapered, extending processes that branch and penetrate some distance into the adjacent neuropil (N). Other areas of ependyma are responsible for production of CSF. X100. H&E. (b) Ependymal cells (E) lining the central canal (C) of the spinal cord help move CSF in that CNS region. X200. H&E.

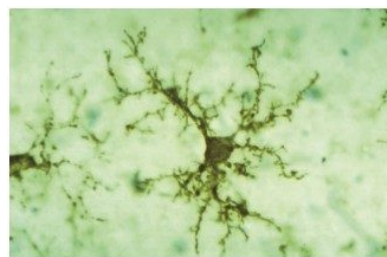
>> MEDICAL APPLICATION

Levels of neurotransmitters in the synaptic cleft and available for binding postsynaptic receptors are normally regulated by several local mechanisms. **Selective serotonin reuptake inhibitors (SSRIs)**, a widely used class of drugs for treatment of depression and anxiety disorders, were designed to augment levels of this neurotransmitter at the postsynaptic membrane of serotonergic CNS synapses by specifically inhibiting its reuptake at the presynaptic membrane.



>> MEDICAL APPLICATION

Alzheimer disease, a common type of dementia in the elderly, affects both neuronal perikarya and synapses within the cerebrum. Functional defects are due to **neurofibrillary tangles**, which are accumulations of tau protein associated with microtubules of the neuronal perikaryon and axon hillock regions, and **neuritic plaques**, which are dense aggregates of β -amyloid protein that form around the outside of these neuronal regions.

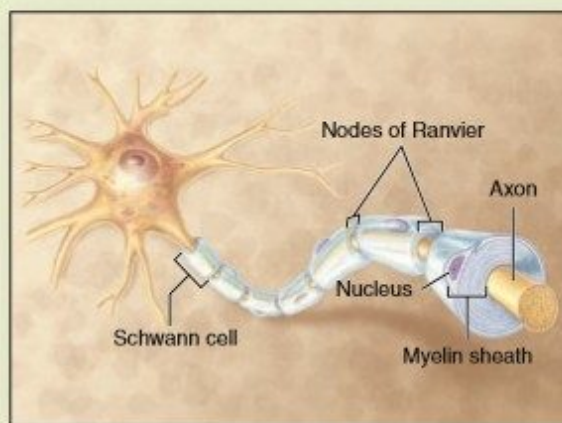


Microglia are monocyte-derived, antigen-presenting cells of the CNS, less numerous than astrocytes but nearly as common as neurons and evenly distributed in both gray and white matter. By immunohistochemistry, here using a monoclonal antibody against human leukocyte antigens (HLA) of immune-related cells, the short branching processes of microglia can be seen. Routine staining demonstrates only the small dark nuclei of the cells. Unlike other glia of the CNS, microglia are not interconnected; they are motile cells, constantly used in immune surveillance of CNS tissues. When activated by products of cell damage or by invading microorganisms, the cells retract their processes, begin phagocytosing the damage- or danger-related material, and behave as antigen-presenting cells. X500. Antibody against HLA-DR and peroxidase.

>> MEDICAL APPLICATION

In **multiple sclerosis (MS)** the myelin sheaths surrounding axons are damaged by an autoimmune mechanism that interferes with the activity of the affected neurons and produces various neurologic problems. T lymphocytes and microglia, which phagocytose and degrade myelin debris, play major roles in progression of this disease. In MS destructive actions of these cells exceeds the capacity of oligodendrocytes to produce myelin and repair the myelin sheaths.

PNS Glial Cells



e Schwann cells

Cell Types

- **Neurons** — the functional/excitable unit; structure: cell body (soma/perikaryon, contains Nissl bodies = RER + free ribosomes, seen as basophilic clumps and most abundant in large cells like motor neurons), dendrites (receive signals via dendritic spines), axon (conducts impulses away from the axon hillock, ending in a terminal bouton at a synapse).
- **Neuroglia (glial cells)** — outnumber neurons roughly 10:1 in the brain; support, nourish, protect, and insulate neurons; do not conduct impulses themselves.

Glial Cell	Location	Function
Astrocytes	CNS	Support neurons, form blood-brain barrier (perivascular feet), regulate extracellular environment; communicate via gap junctions
Oligodendrocytes	CNS	Myelinate multiple CNS axons (one cell wraps several axons)
Microglia	CNS	Immune defence — derived from blood monocytes; phagocytose debris/pathogens
Ependymal cells	CNS (line ventricles)	Produce/circulate cerebrospinal fluid (CSF); ciliated, involved in CSF flow
Schwann cells	PNS	Myelinate ONE PNS axon segment each; aid nerve regeneration
Satellite cells	PNS	Support/insulate neuron cell bodies in ganglia

Neuron Classification

- By structure: **multipolar** (many dendrites, one axon — most common, e.g. motor neurons), **bipolar** (one dendrite, one axon — retina, olfactory, cochlear/vestibular ganglia), **unipolar/pseudounipolar** (single process splits — sensory/spinal ganglia neurons), **anaxonic** (many dendrites, no true axon — modulate nearby neurons).
- By function: **sensory (afferent)** — carry impulses toward CNS; **motor (efferent)** — carry impulses away from CNS to effectors (somatic = voluntary skeletal muscle, autonomic = involuntary glands/cardiac/smooth muscle); **interneurons** — relay/process signals within the CNS (make up an estimated 99% of CNS neurons).

Nerve Impulse & Synapse

- Resting membrane potential is about -65 mV (inside negative), maintained by the Na^+/K^+ pump. A threshold stimulus opens voltage-gated Na^+ channels, depolarizing the membrane to about $+30$ mV; K^+ channels then open to repolarize it — this self-propagating wave is the action potential/nerve impulse.
- Axoplasmic transport is bidirectional: anterograde transport carries materials from soma to synaptic terminals; retrograde transport carries materials (including some viruses/toxins) back toward the cell body.
- A **synapse** converts an electrical signal into a chemical one via neurotransmitter release across a synaptic cleft; types include axosomatic, axodendritic, and axoaxonic. Major neurotransmitter classes: catecholamines (adrenaline, noradrenaline, dopamine), amino acids (glutamate, glycine, serotonin, GABA), and small peptides (endorphins, substance P). Acetylcholine is used at the neuromuscular junction.

■ EXAM HACK — Myelination trap

CNS myelin = Oligodendrocytes (one cell, MANY axons). PNS myelin = Schwann cells (one cell, ONE axon segment). Mnemonic: 'Oligo' sounds like 'many' relationships at once; 'Schwann' = one-on-one. Multiple sclerosis attacks CNS myelin (oligodendrocytes); Guillain-Barré attacks PNS myelin (Schwann cells).

8. Blood

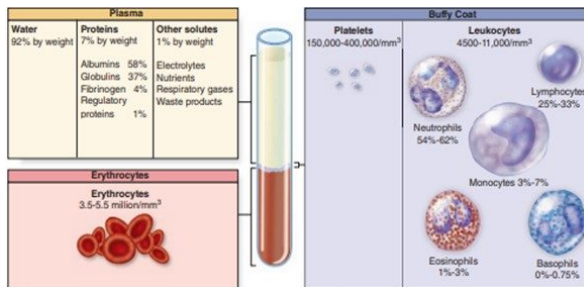
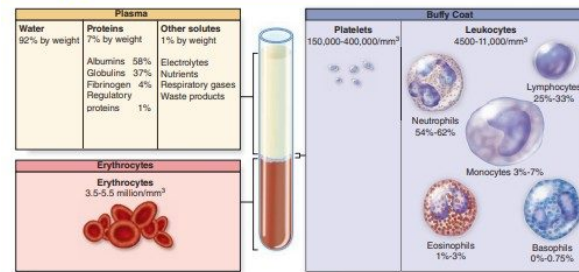


FIGURE 12-1 Composition of whole blood.



A tube of blood after centrifugation (center) has 36%-53% of its volume represented by erythrocytes in the bottom half of the tube, a volume called the **hematocrit**. Between the sedimented erythrocytes and the supernatant light-colored plasma is a thin layer of leukocytes and platelets called the **buffy coat**. The concentration ranges of erythrocytes, platelets, and leukocytes in normal blood are included here, along with the **differential count** or percent range for each type of leukocyte represented in the buffy coat. A cubic millimeter of blood is equivalent to a microliter (μL). [All complete blood count (CBC) values in this chapter are those used by the US National Board of Medical Examiners.]

FIGURE 12-3 Preparing a blood smear.

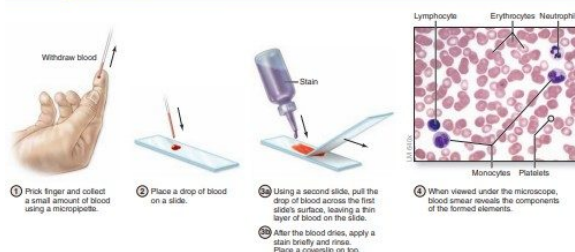
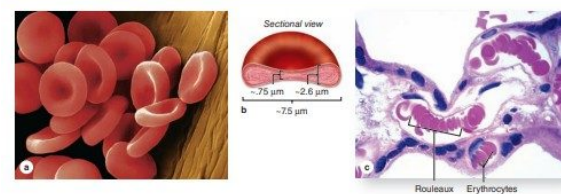


FIGURE 12-4 Normal human erythrocytes.



>> MEDICAL APPLICATION

Anemia is the condition of having a concentration of erythrocytes below the normal range. With fewer RBCs per milliliter of blood, tissues are unable to receive adequate O₂. Symptoms of anemia include lethargy, shortness of breath, fatigue, skin pallor, and heart palpitations. Anemia may result from insufficient red cell production, due, for example, to iron deficiency, or from blood loss with a stomach ulcer or excessive menses.

An increased concentration of erythrocytes in blood (**erythrocytosis**, or **polycythemia**) may be a physiologic adaptation found, for example, in individuals who live at high altitudes, where O₂ tension is low. Elevated hematocrit increases blood viscosity, putting strain on the heart, and, if severe, can impair circulation through the capillaries.

>> MEDICAL APPLICATION

Inherited alterations in hemoglobin molecules are responsible for several pathologic conditions, an example of which is **sickle cell disease**. This disorder is caused by a mutation of one nucleotide (a **point mutation**) in the gene for the hemoglobin β chain. The consequences of this single substitution are profound. When the altered hemoglobin (called **HbS**) is deoxygenated in capillaries, it polymerizes and forms rigid aggregates that cause a characteristic sickle shape (Figure 12-5). The sickled erythrocyte is less flexible and more fragile and has a shortened life span that can lead to anemia. It increases the blood viscosity and can damage the wall of blood vessels, promoting blood coagulation. Sick cells can block capillaries, restricting O₂ delivery to tissues and leading to varying degrees of ischemia or **anoxia** and organ damage.

FIGURE 12-7 Neutrophils.

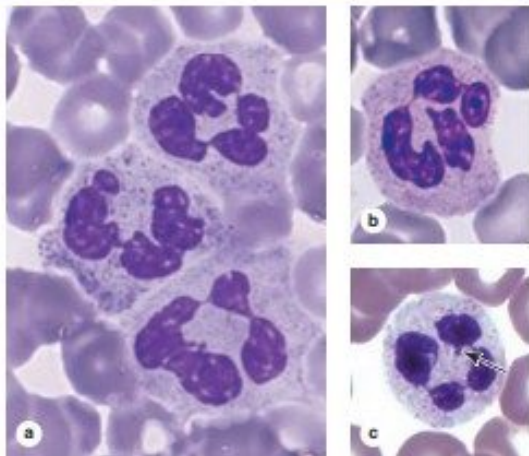
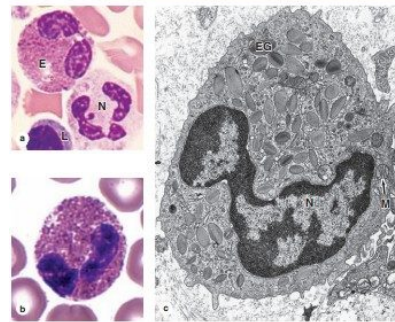
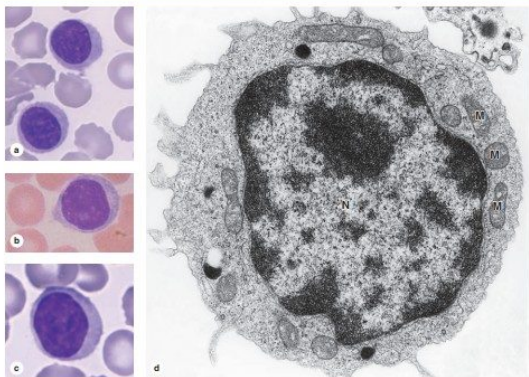


FIGURE 12-8 Eosinophils.



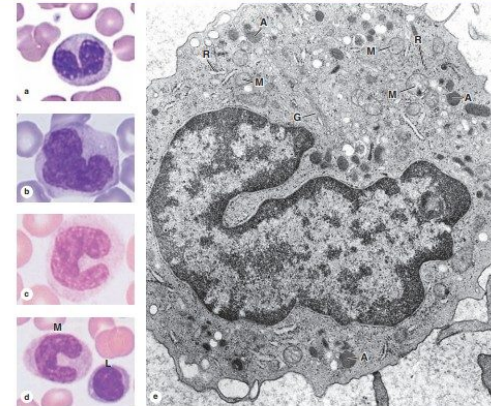
Eosinophils are about the same size as neutrophils but have bilobed nuclei and more abundant coarse cytoplasmic granules. The cytoplasm is often filled with brightly eosinophilic specific granules, but it also includes some azurophilic granules. (a) Micrograph shows an eosinophil (E) next to a neutrophil (N) and a small lymphocyte (L). X1500, Wright. (b) Even with granules filling the cytoplasm, the two nuclear lobes of eosinophils are usually clear. X1500, Giemsa. (c) Ultrastructurally a sectioned eosinophil clearly shows the unique specific eosinophilic granules (EG), as oval structures with dense, electron-dense, crystalline cores. These granules, along with a few lysosomes and mitochondria (M), fill the cytoplasm around the bilobed nucleus (N). X20,000.

FIGURE 12-9 Lymphocytes.



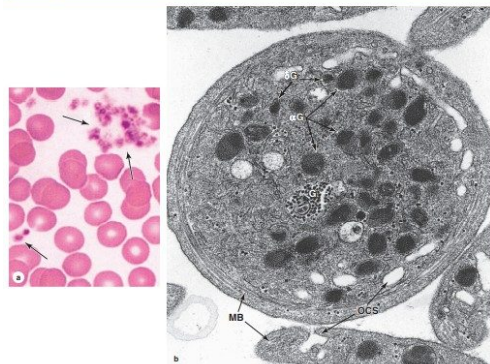
Lymphocytes are agranulocytes and lack the specific granules characteristic of granulocytes. Lymphocytes circulating in blood generally range in size from 6 to 15 μm in diameter and are sometimes classified arbitrarily as small, medium, and large. (a) The most numerous small lymphocytes shown here are slightly larger than the neighboring erythrocytes and have only a thin rim of cytoplasm surrounding the spherical nucleus. X1500, Giemsa. (b) Medium lymphocytes are distinctly larger than erythrocytes. X1500, Wright. (c) Large lymphocytes, much larger than erythrocytes, may represent activated cells that have returned to the circulation. X1500, Giemsa. (d) Ultrastructurally a medium-sized lymphocyte is seen to be mostly filled with a euchromatic nucleus (N) surrounded by cytoplasm containing mitochondria (M), free polyosomes, and a few dark lysosomes (azurophilic granules). X22,000.

FIGURE 12-10 Monocytes.

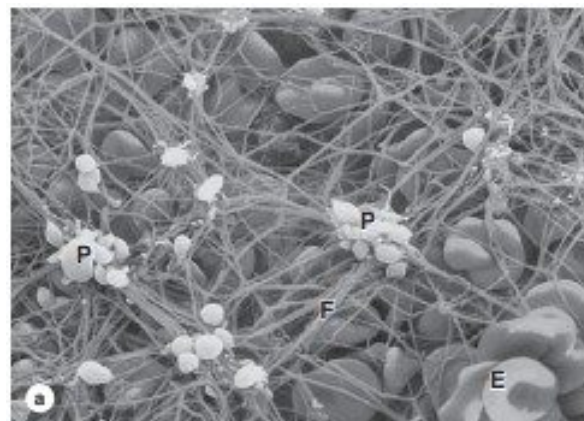


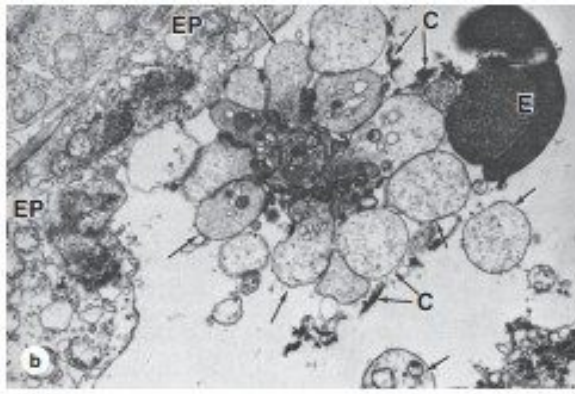
Monocytes are large agranulocytes with diameters from 12 to 20 μm that circulate as precursors to macrophages and other cells of the mononuclear phagocyte system. (a-d) Micrographs of monocytes showing their distinctive nuclei which are indented, kidney-shaped, or C-shaped. a: X1500, Giemsa; b-d: X1500, Wright. (e) Ultrastructurally the cytoplasm of a monocyte shows a Golgi apparatus (G), mitochondria (M), and lysosomes or azurophilic granules (A). Rough ER is poorly developed and there are some free polyosomes (P). X22,000. (Figure 12-12e, with permission, from D.F. Bainton and M.G. Farquhar, Department of Pathology, University of California at San Francisco, CA.)

FIGURE 12-11 Platelets.



Platelets are cell fragments 2-4 μm in diameter derived from megakaryocytes of bone marrow. Their primary function is to rapidly release the content of their granules upon contact with collagen (or other materials outside of the endothelium) to begin the process of clot formation and reduce blood loss from the vasculature. (a) In a blood smear, platelets (arrows) are often found as aggregates. Individually they show a lightly stained **hyalomere** region surrounding a more darkly stained central **granulomere** containing membrane-enclosed granules. X1500, Wright. (b) Ultrastructurally a platelet shows a system of microtubules and actin filaments near the periphery, called the **marginal bundle (MB)**, which is formed as the platelet pinches off from megakaryocytes (Chapter 13), and helps maintain its shape. An open circulatory system (OCS) of invaginating membrane vesicles continuous with the plasmalemma facilitates rapid degradation upon activation and Ca^{2+} release. The central granulomere region contains small dense delta granules (DG), larger and more numerous alpha granules (AG), and glycogen (G). X30,000. (Figure 12-23b, with permission, from Dr M. J. G. Harrison, Middlesex Hospital and University College London, UK.)





Overview

- Classified as a fluid connective tissue — cells (formed elements) suspended in a liquid matrix (plasma); about 5 L circulates in an average adult.
- **Plasma** (~55%): water, plasma proteins (albumin — maintains osmotic pressure; globulins — transport factors, immunoglobulins; fibrinogen — clotting), electrolytes, nutrients, wastes, hormones.
- **Formed elements** (~45% = the haematocrit): erythrocytes, leukocytes, platelets — all arise from haematopoietic stem cells in red bone marrow. The thin buffy coat between plasma and packed RBCs holds leukocytes and platelets.

Formed Elements

Cell	Key Feature	Function
Erythrocytes (RBCs)	Biconcave disc, no nucleus, no mitochondria, packed with haemoglobin; survive ~120 days	Transport O ₂ (as oxyhaemoglobin) and CO ₂ (as carbaminohaemoglobin/HCO ₃ ⁻)
Neutrophils	Multilobed (2-5 lobe) nucleus, granular cytoplasm; most abundant WBC (54-62%)	First responders — phagocytose bacteria; azurophilic granules (myeloperoxidase, lysozyme, defensins) and specific granules kill/digest microbes
Eosinophils	Bilobed nucleus, red/orange granules (1-3% of WBCs)	Combat parasites (major basic protein) and modulate allergic reactions
Basophils	Large dark granules often obscuring the nucleus; least common WBC (<1%)	Release histamine/heparin (inflammation); central to type I hypersensitivity along with mast cells
Lymphocytes	Large nucleus, thin rim of cytoplasm; most numerous agranulocyte	B cells (antibodies), T cells (CD4+ helper/CD8+ cytotoxic, cell-mediated immunity), NK cells
Monocytes	Largest WBC, kidney/C-shaped nucleus	Circulate then differentiate into macrophages, osteoclasts, microglia — phagocytosis and antigen presentation
Platelets (thrombocytes)	Cell fragments, no nucleus, 2-4 µm, from megakaryocytes; lifespan ~10 days	Blood clotting (haemostasis)

Platelet Function in Haemostasis

- **Primary aggregation:** platelet glycocalyx adheres to exposed collagen at a vessel injury, forming an initial plug.
- **Secondary aggregation:** platelets release ADP and adhesive glycoproteins, recruiting more platelets and enlarging the plug.
- **Coagulation:** fibrinogen and other factors trigger a cascade producing cross-linked fibrin, trapping RBCs and platelets into a clot (thrombus).
- **Clot retraction** (platelet actin-myosin) then **clot removal** by the enzyme plasmin once the vessel wall is repaired.

■ EXAM HACK — WBC recall order

'Never Let Monkeys Eat Bananas' = Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils = descending order of abundance. Trap: platelets are NOT true cells — they're cytoplasmic fragments of megakaryocytes, and have no nucleus.

9. Cartilage

General Characteristics

- A specialized, resilient, semi-rigid connective tissue that resists compression while allowing some flexibility — the firm matrix (chondroitin sulphate) gives cartilage its consistency.
- **Avascular** (except the perichondrium) and has no nerves of its own; nourished by diffusion, heals slowly.
- Cells: **chondroblasts** in the inner perichondrium synthesize type II collagen and matrix; once surrounded by matrix they mature into **chondrocytes**, housed singly or in groups (a 'cell nest') in small cavities called **lacunae**.
- Surrounded (except at joint surfaces) by a fibrous **perichondrium**, which supplies nutrients and allows appositional growth.
- Growth is by two mechanisms: **interstitial growth** — chondrocytes proliferate and lay down new matrix from within (only in young, malleable cartilage); **appositional growth** — new layers are added to the surface from the perichondrium.

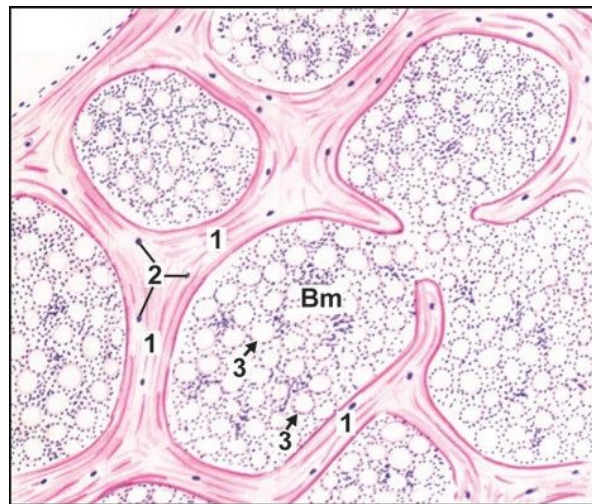
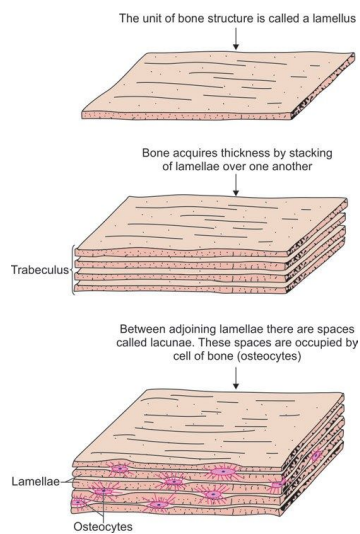
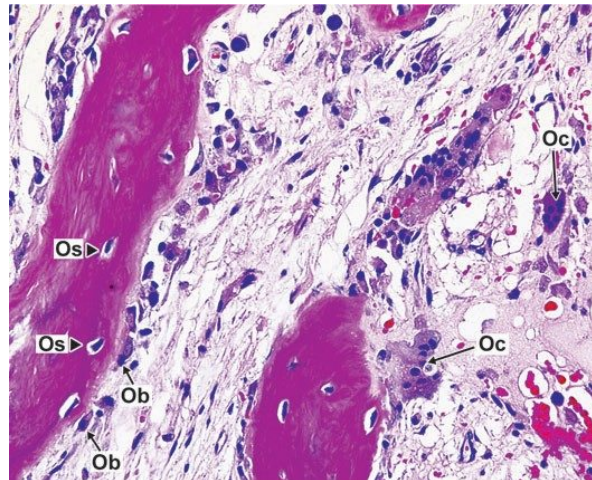
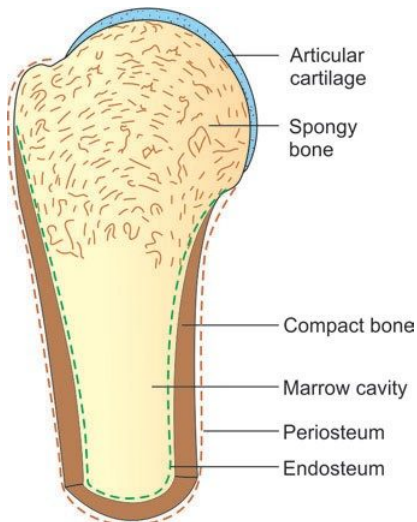
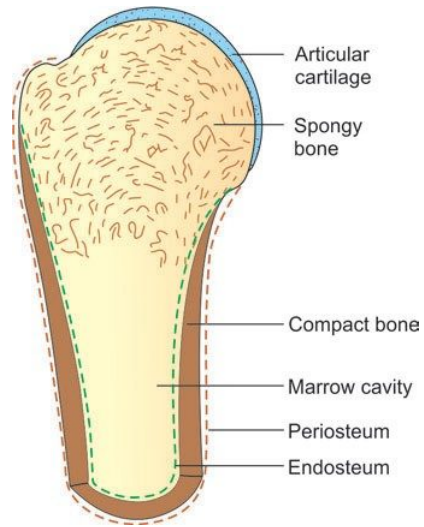
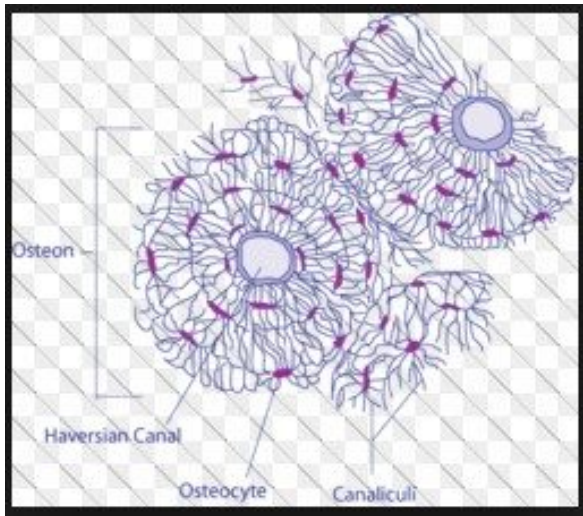
Types

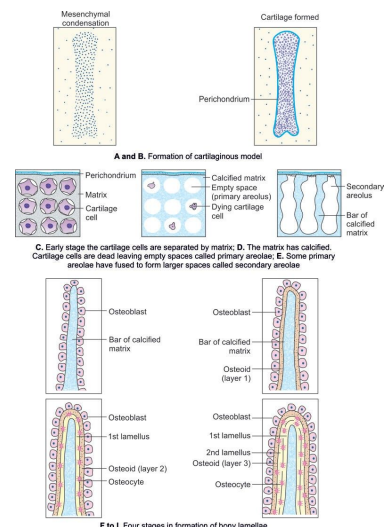
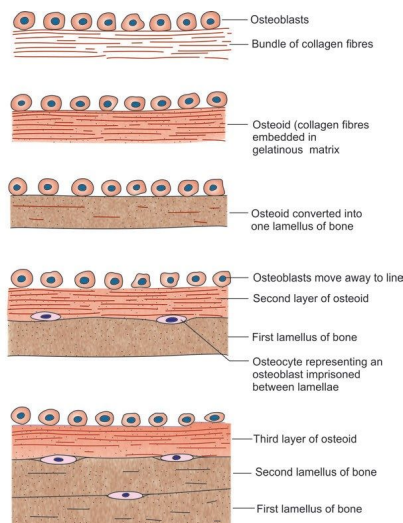
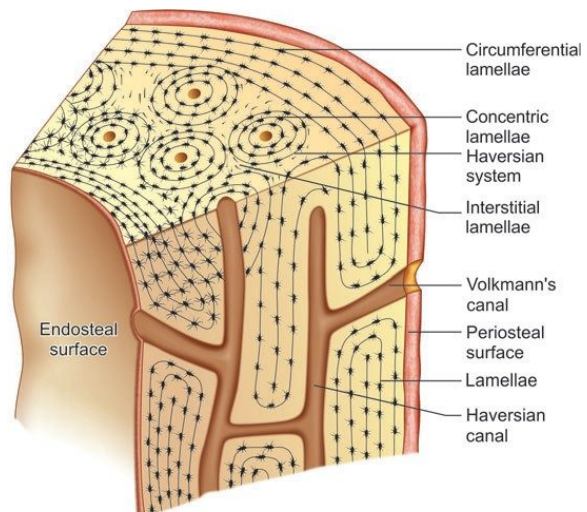
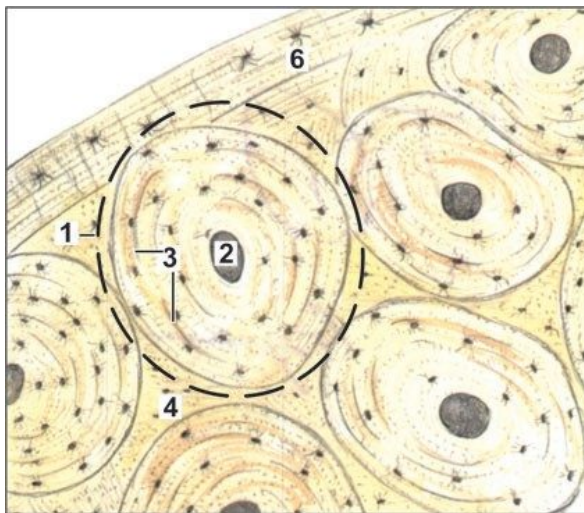
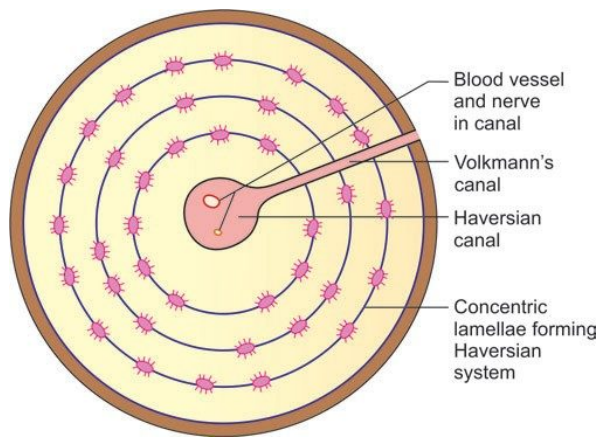
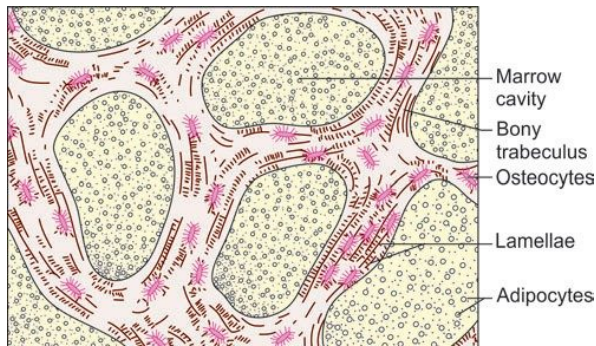
Type	Matrix Features	Location	Notes
Hyaline	Glassy, smooth, fine type II collagen fibres	Articular surfaces, trachea, nasal septum, costal cartilage, fetal skeleton	Most common type; forms most of the fetal skeleton before endochondral ossification replaces it with bone
Elastic	Dense elastic fibre network instead of collagen	External ear (pinna), epiglottis, auditory tube, laryngeal cartilages	Provides flexible support; recovers shape after deformation
Fibrocartilage	Thick bundles of visible collagen fibres, chondrocytes in short columns	Intervertebral discs, pubic symphysis, menisci	Great tensile strength + elasticity + rigidity; NEVER has a perichondrium

■ EXAM HACK — Cartilage ID trap

Hyaline = 'glassy/glass-like' — the DEFAULT answer whenever a question just says 'cartilage' without more detail. Fibrocartilage NEVER has a perichondrium — it blends directly into surrounding dense connective tissue, a favourite trick point.

10. Bone





General Characteristics

- A specialized, rigid connective tissue — matrix is calcified (hydroxyapatite crystals: calcium + phosphate) giving it hardness. Bone composition (living bone): ~20% water, 30-40% organic material (mainly type I collagen — 'osteoid collagen'), 40-50% inorganic mineral salts.

- Highly **vascularized** and capable of self-repair, unlike cartilage.
- Functions: support, protection, movement (lever for muscles), mineral storage (Ca²⁺, PO₄³⁻, ~99% of body calcium), and haematopoiesis (red marrow).
- **Periosteum** covers the outer surface (except at joints) — outer fibrous layer + inner osteogenic/cellular layer rich in osteoprogenitor cells; provides attachment for tendons/ligaments and mediates appositional growth and fracture repair. **Endosteum** lines the internal marrow cavity.

Bone Cells

Cell	Function
Osteoprogenitor cells	Mesenchyme-derived stem cells lining periosteum/endosteum; differentiate into osteoblasts
Osteoblasts	Bone-BUILDING cells — cuboidal/columnar, rich in alkaline phosphatase; synthesize osteoid (unmineralized matrix) and initiate its mineralization
Osteocytes	Mature osteoblasts trapped in lacunae — maintain bone matrix, sense mechanical stress, help regulate calcium exchange via canaliculi
Osteoclasts	Large, multinucleate (up to 20+ nuclei), bone-RESORBING cells occupying Howship's lacunae; rich in acid phosphatase and mitochondria; derived from the monocyte lineage

Microscopic Structure

- **Compact (cortical) bone:** dense, organized into structural units called **osteons (Haversian systems)** — concentric lamellae around a central (Haversian) canal carrying blood vessels/nerves; Volkmann's canals connect adjacent osteons and communicate with the periosteum and marrow cavity. Adapted to resist bending/twisting forces.
- **Spongy (cancellous) bone:** lattice of thin plates called **trabeculae**, with spaces filled by red bone marrow — lighter, found at the ends of long bones and within flat bones; no true osteons, adapted to resist compressive forces.
- Long bones: elongated shaft (diaphysis) + two expanded articular ends (epiphyses). **Sesamoid bones** (e.g. patella) are embedded within tendons/joint capsules, resisting pressure, reducing friction, and altering the direction of muscle pull.

Bone Formation (Ossification)

- **Intramembranous ossification:** bone forms directly from mesenchyme (no cartilage model) — flat bones of the skull, mandible, clavicle. Mesenchymal condensation → vascularization → osteoblasts secrete osteoid → calcium salts deposit to form the first lamellus.
- **Endochondral ossification:** a hyaline cartilage model is gradually replaced by bone — most bones of the body, including long bones (growth continues at the epiphyseal/growth plate until maturity).
- Growth in bone occurs only by **apposition** (deposition of new bone on existing bone), never interstitially like cartilage.

■ ALZHEIMER'S MODE

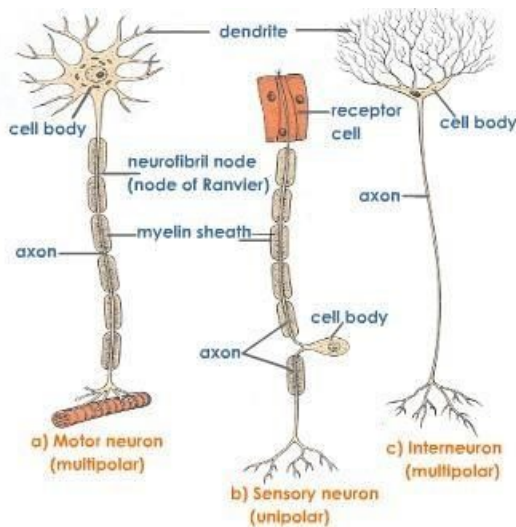
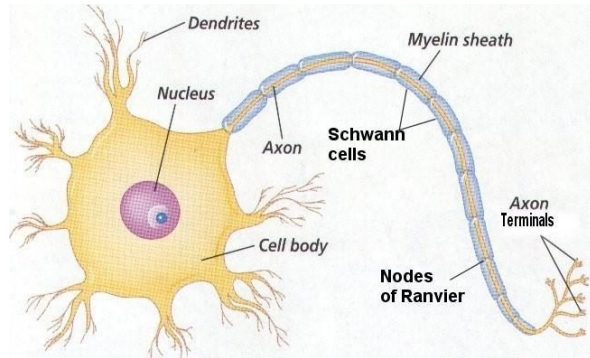
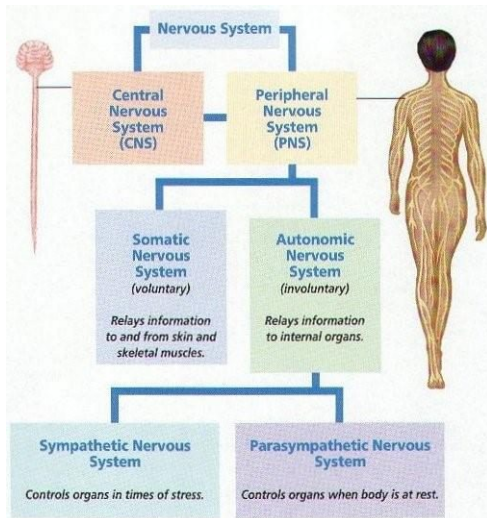
Osteoblast = 'B' for Build. Osteoclast = 'C' for Chew/Chomp (breaks bone down). The osteon is like a tree trunk's growth rings — concentric layers (lamellae) wrapped around a central pipe (Haversian canal) carrying the blood supply.

■ **EXAM HACK — Bone trap**

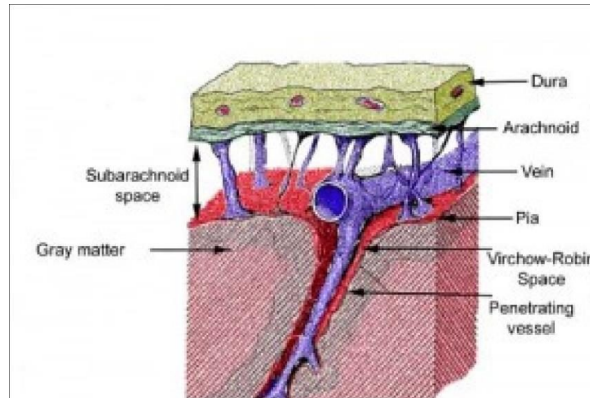
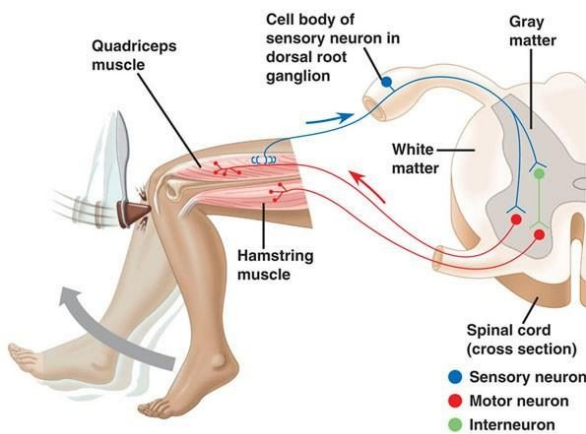
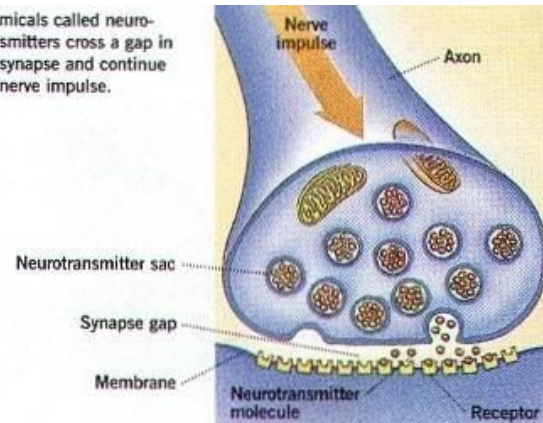
Osteocytes are TRAPPED, mature, maintenance-only cells — they do not divide or actively build new matrix (osteoblasts do that). Growth plate (epiphyseal plate) is CARTILAGE, not bone — a classic 'which of these is NOT ossified' trick question.

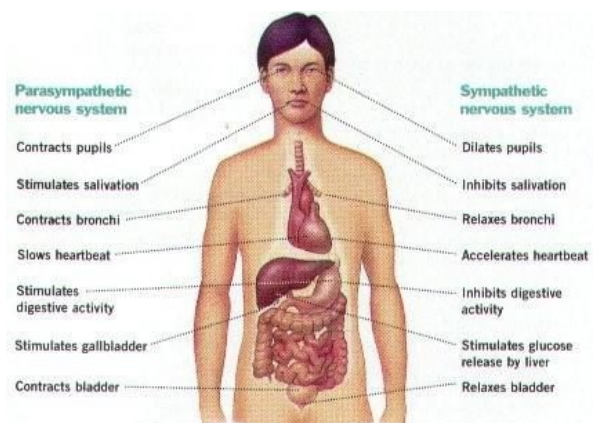
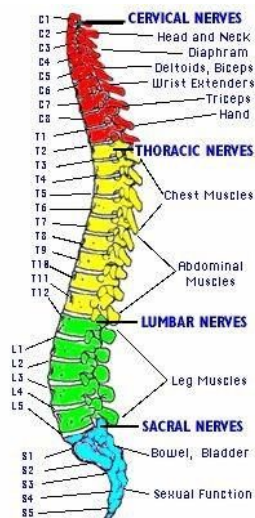
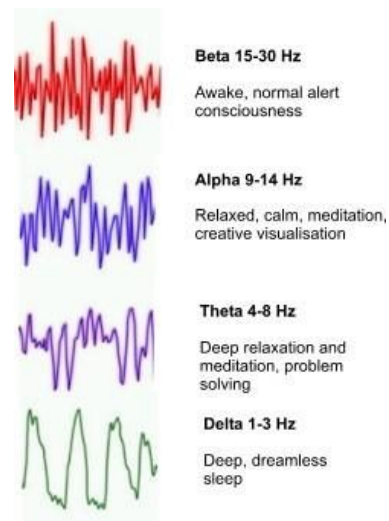
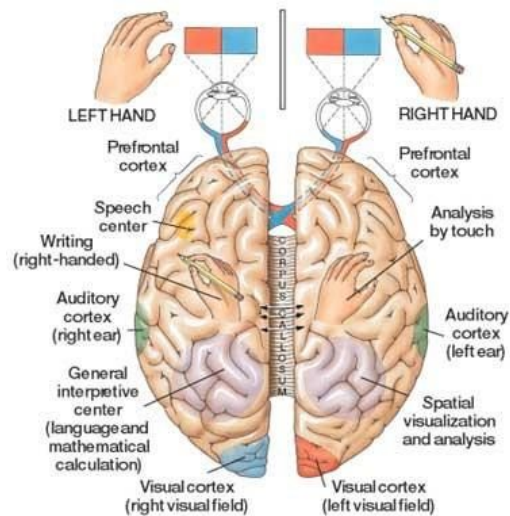
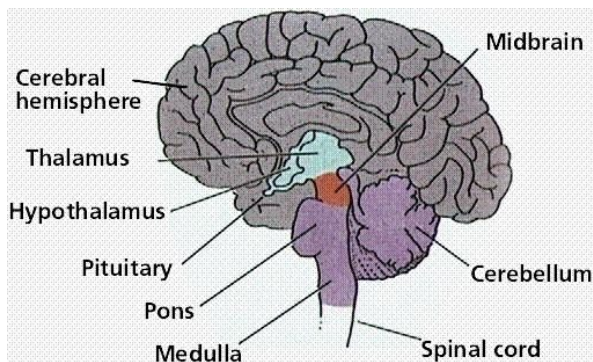
11. Bonus — Nervous System Overview

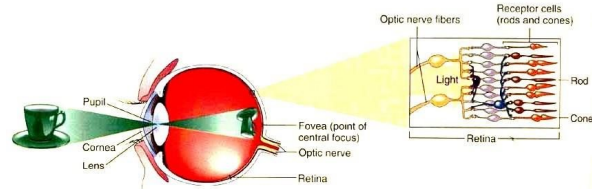
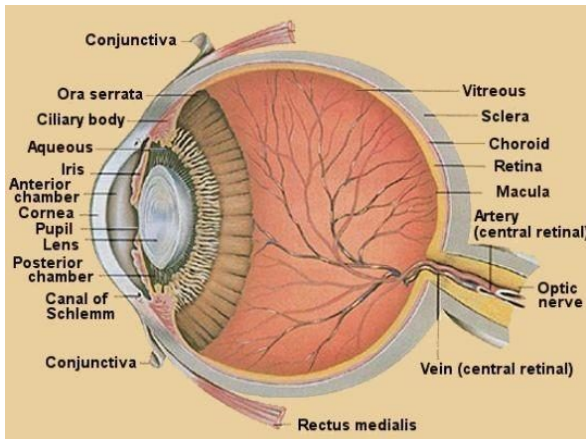
Pulled in from the course's Nervous System handout to give histology-level cell types (Section 7) their whole-organ context. Not a replacement for a dedicated Physiology/Neuroanatomy pass — use this as a fast orientation map.



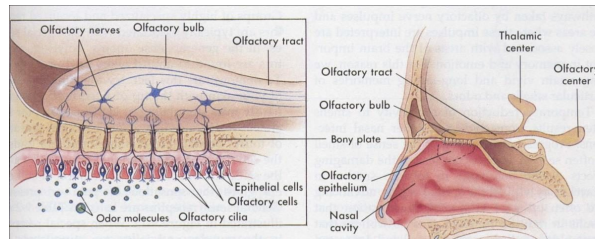
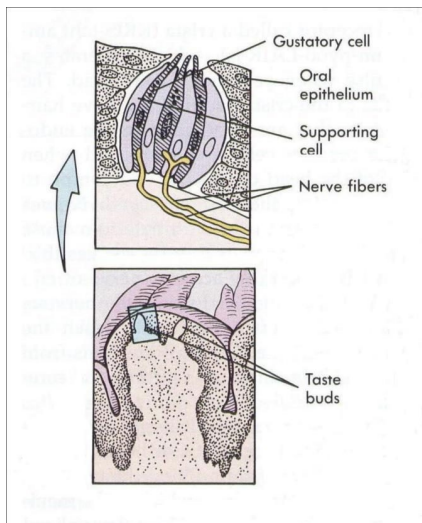
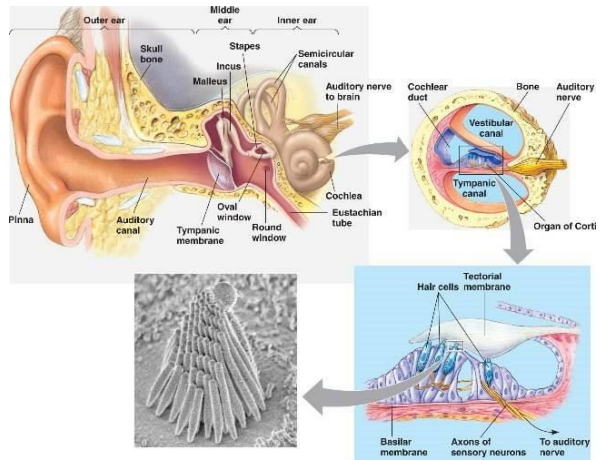
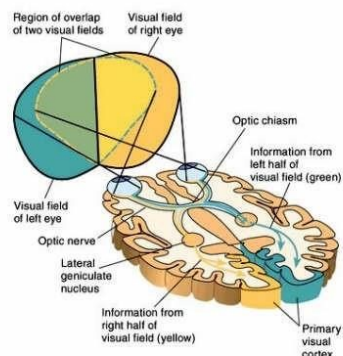
Chemicals called neurotransmitters cross a gap in the synapse and continue the nerve impulse.

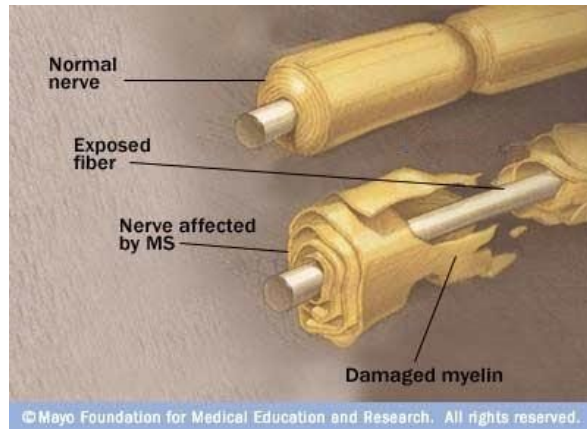
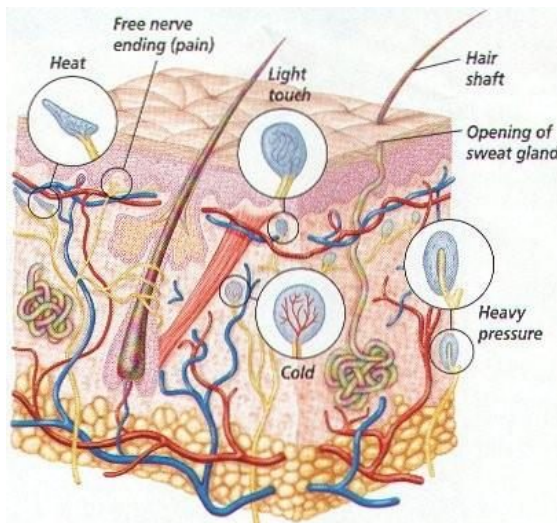






► The Primary Visual Pathway





Divisions & Functions

- **Functions:** gathers information from inside/outside the body (sensory), transmits it to processing centres (conduction), processes it (integration), and sends signals to muscles/glands/organs (motor) — maintaining homeostasis and coordinating all other body systems.
- **Central Nervous System (CNS):** brain + spinal cord. **Peripheral Nervous System (PNS):** 12 pairs of cranial nerves + 31 pairs of spinal nerves, plus ganglia.
- **Somatic nervous system** (voluntary): relays skin/sense-organ/skeletal-muscle information to the CNS and carries voluntary motor responses back. **Autonomic nervous system** (involuntary): regulates internal organs via sympathetic ('fight or flight', stress response) and parasympathetic ('rest and digest', normal conditions) divisions.

Meninges & CSF

- Three protective coverings around brain and spinal cord: **dura mater** (tough, outermost), **arachnoid mater** (middle, web-like), **pia mater** (thin, transparent, follows every brain contour).
- Cerebrospinal fluid flows through the subarachnoid space (between arachnoid and pia), buffering, nourishing, and detoxifying the CNS.

Regions of the Brain

Region	Function
Cerebrum	Largest part (~2/3 of brain mass); conscious activity — perception, emotion, thought, planning. Cortex (gray matter, outer 1.5-5mm) is folded into gyri/sulci to maximise surface area; frontal lobe = movement/planning, parietal = sensory processing, temporal = hearing/speech, occipital = vision
Cerebellum	Coordination of movement and motor learning
Diencephalon	Thalamus = sensory relay/'switchboard'; hypothalamus = regulates internal organs, pituitary hormones, sleep and appetite
Brainstem (medulla, pons, midbrain)	Involuntary reflex centres — cardiac, vasomotor and respiratory centres in the medulla; breathing centres in the pons; visual/auditory reflex centres in the midbrain

Impulses, Reflex Arc & Major Sense Organs

- A stimulus strong enough to trigger a response is converted by an excitable neuron into a nerve impulse (all-or-nothing rule); impulses are self-propagating and unidirectional (dendrite → cell body → axon).
- **Reflex arc:** Receptor → afferent (sensory) pathway → interneuron/CNS synapse → efferent (motor) pathway → effector (muscle/gland). Spinal reflexes complete at the spinal cord without higher brain involvement.
- **Eye:** three layers — outer (sclera/cornea), middle (choroid, ciliary body, iris), inner (retina). Rods = dim-light/peripheral vision; cones (blue/red/green) = colour/detail vision, concentrated at the fovea. Optic nerves cross at the optic chiasm before reaching the visual cortex.
- **Ear:** outer ear/canal → eardrum → middle-ear ossicles (hammer, anvil, stirrup) amplify vibration → cochlea converts vibration to nerve signals via the auditory nerve; semicircular canals provide balance/spatial orientation.
- **Taste/smell:** taste buds detect five tastes (sweet, sour, bitter, salty, umami); olfactory receptors in the nasal roof detect airborne chemicals — closely linked to the limbic system (hence smell-triggered memories).
- **Skin:** Meissner's corpuscles (light touch), Pacinian corpuscles (deep pressure/vibration), free nerve endings (pain — most numerous), and separate thermoreceptors for cold/warmth.

Common Disorders (orientation only)

- **Epilepsy** — chronic disorder of recurrent seizures from abnormal electrical discharges in the brain.
- **Alzheimer's disease** — degenerative dementia with progressive memory/judgment loss.
- **Multiple sclerosis** — autoimmune destruction of CNS myelin (oligodendrocytes), affecting vision, sensation, coordination.
- **Parkinson's disease** — low brain dopamine, causing tremor and movement/coordination difficulty.

■ End of Compendium

This expanded edition consolidates every ANA203 topic taught to date — The Cell, Membrane Transport, Cell Division, Epithelial Tissue, Connective Tissue Proper, Muscle Tissue, Nerve Tissue, Blood, Cartilage, and Bone — enriched with deeper mechanism-level detail, plus a bonus Nervous System overview. Pair this with the ANA203 CBTs on the INDIGENE Study Console for active recall practice.